Quantum Noise Simulation

Assessment of noise models, channels, simulation and mitigation

ANSWER BOOK

This book accompanies the separate *Question Paper* (80 questions, 3 hours, 240 marks). It contains

1. the **answer key**, section by section, with the topic tested by every question (page 2),
2. a **topic-wise map** grouping the questions by theme and a **quick-marking grid** (page 3), and
3. the **Answer Bank**: every question restated with the correct option marked ✓ and a complete solution with derivations (from page 4).

Marking scheme (every question)

Each question has **four** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Full marks:	+3 if ONLY the correct option is chosen;
Zero marks:	0 if none of the options is chosen (question unanswered);
Negative marks:	−1 in all other cases (wrong option, or more than one option chosen).

How to evaluate

For each section count N_c = number of correct answers and N_w = number of questions answered wrongly (or with more than one bubble darkened). Section marks = $3N_c - N_w$; unanswered questions score 0. The total is out of 240. Every numerical answer was re-computed independently (NumPy Kraus-operator and density-matrix calculations, exact arithmetic where possible, and CUDA-Q density-matrix and stabilizer simulations, including the noise model of `Noise_Simulation.ipynb`) by `verify_answers.py` in the source folder.

Conventions (identical to the Question Paper)

- Qubit 0 is written first: $|q_0 q_1 \ldots\rangle$; two-qubit matrices use the basis order $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. X, Y, Z are the Pauli matrices, H the Hadamard gate, $R_P(\theta) = e^{-i\theta P/2}$; CNOT flips the target when the control is 1.
- A qubit state is $\rho = \frac{1}{2}(I + r_x X + r_y Y + r_z Z)$ with Bloch vector $\vec{r}$, $|\vec{r}| \le 1$; purity $\mathrm{Tr}\,\rho^2$. State fidelity with a pure target: $F = \langle\psi|\rho|\psi\rangle$. Measurement probabilities are the diagonal of ρ; $\langle Z\rangle = P(0) - P(1)$.
- A channel is $\mathcal{E}(\rho) = \sum_k K_k \rho K_k^\dagger$ with $\sum_k K_k^\dagger K_k = I$. Bit flip: $\sqrt{1-p}\,I, \sqrt{p}\,X$; phase flip: $\sqrt{1-p}\,I, \sqrt{p}\,Z$; bit-phase flip: $\sqrt{1-p}\,I, \sqrt{p}\,Y$; depolarizing (CUDA-Q): $\sqrt{1-p}\,I, \sqrt{p/3}\,X, \sqrt{p/3}\,Y, \sqrt{p/3}\,Z$ (two-qubit: $\sqrt{1-p}\,II$ and the 15 other Paulis with $p/15$); amplitude damping: $\mathrm{diag}(1, \sqrt{1-\gamma})$, $\sqrt{\gamma}\,|0\rangle\langle 1|$; phase damping: $\mathrm{diag}(1, \sqrt{1-\lambda})$, $\mathrm{diag}(0, \sqrt{\lambda})$.
- T_1: $\rho_{11}(t) = \rho_{11}(0)e^{-t/T_1}$; T_2: $|\rho_{01}(t)| = |\rho_{01}(0)|e^{-t/T_2}$; $\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_\varphi}$. ln is the natural logarithm.
- **Notebook** (`Noise_Simulation.ipynb`, defaults): 15-qubit GHZ circuit (H on qubit 0, CNOT chain), the chosen channel with probability p after every `h`, `x`, `ry`, `rz` gate and a two-qubit channel after every CNOT; no measurement noise; 10 000 shots, seed 42. **Previously recorded run**: CUDA-Q 0.16.0, CPU target `stim`, no GPU, depolarizing $p = 0.01$: population 0.8936, X-basis parity 0.6910, $(1-p)^G = 0.8601$. No physical quantum processor was used.

Answer key

Answer distribution: (A) 20, (B) 20, (C) 20, (D) 20.

Section A — Mathematical and Quantum Foundations

Q.	Ans.	Topic tested	Q.	Ans.	Topic tested
1	**C**	Density matrix of a pure state	11	**C**	Partial trace; entanglement and mixedness
2	**C**	Purity $\mathrm{Tr}\,\rho^2$	12	**B**	Eigenvalues of a qubit ρ
3	**B**	Bloch vector from ρ	13	**A**	Independent qubits: tensor product
4	**D**	Validity of density matrices	14	**C**	Conjugation $X\rho X$ on the Bloch sphere
5	**A**	Trace preservation of Kraus sets	15	**D**	Bloch vectors of states
6	**C**	Random phases $\rightarrow$ dephasing	16	**D**	Hermitian, unitary, projector, Kraus
7	**A**	Error-free probability $(1-p)^n$	17	**A**	Mixed states inside the Bloch ball
8	**B**	Shot-noise standard error	18	**D**	Density-matrix conditions; Born rule
9	**A**	Coherent angle error	19	**D**	Tomography: Bloch vector, purity
10	**B**	Fidelity with a mixed state	20	**B**	Tomography: fidelity, probabilities

Section B — Quantum Noise Fundamentals

Q.	Ans.	Topic tested	Q.	Ans.	Topic tested
21	**D**	Fixed points of bit flip	31	**A**	Readout errors on GHZ
22	**D**	Phase flip on $\lvert+\rangle$	32	**B**	Coherent vs incoherent errors
23	**C**	Depolarizing flip probability	33	**B**	Two-qubit depolarizing marginal
24	**A**	Amplitude damping on $\lvert+\rangle$	34	**A**	Phase errors and the X basis
25	**C**	Phase damping $\equiv$ phase flip	35	**B**	Kraus operators of channels
26	**D**	T_1 half-life	36	**D**	Channels on the Bloch ball
27	**D**	T_2 from T_1 and T_φ	37	**B**	Non-unital amplitude damping
28	**B**	Constraint $T_2 \leq 2T_1$	38	**C**	Complete dephasing vs unitary Z
29	**A**	Coherence-limited depth	39	**A**	T_1/T_2 decay of $\lvert+\rangle$
30	**C**	Readout error probabilities	40	**C**	Fidelity and T_φ

Section C — Noise Models and Simulation

Q.	Ans.	Topic tested	Q.	Ans.	Topic tested
41	**B**	CUDA-Q depolarizing convention	51	**D**	Hellinger fidelity and TVD
42	**A**	Constant identity gates and noise	52	**C**	Parity from X-basis counts
43	**D**	CPTP check	53	**D**	Trajectories vs density matrix
44	**B**	Applying the bit-flip channel	54	**D**	`add_all_qubit_channel`
45	**C**	Composing depolarizing channels	55	**A**	Noise-capable CUDA-Q targets
46	**C**	Notebook: depolarizing $(1-p)^G$	56	**C**	Figures of merit
47	**A**	Notebook: bit-flip $(1-p)^{29}$	57	**D**	$(1-p)^G$: bound, not exact value
48	**A**	Density-matrix memory	58	**A**	Density-matrix path in CUDA-Q 0.16
49	**C**	What `stim` can simulate	59	**B**	Recorded run: leaked probability
50	**B**	`as_counts_from_rho`	60	**B**	Recorded run: population vs parity

Section D — Advanced Noise Analysis and Implementation

Q.	Ans.	Topic tested	Q.	Ans.	Topic tested
61	**A**	Process vs average gate fidelity	71	**B**	Surface-code scaling
62	**D**	Randomized benchmarking	72	**A**	Coherent vs stochastic accumulation
63	**D**	Linear ZNE	73	**B**	Pauli twirling
64	**C**	Richardson ZNE	74	**D**	Notebook: `auto` method choice
65	**C**	Gate folding scale factor	75	**A**	Mitigation, DD, QEC
66	**B**	Readout mitigation	76	**B**	Error taxonomy
67	**D**	Limits of readout mitigation	77	**C**	Cost and nature of ZNE
68	**D**	Ramsey vs Hahn echo	78	**B**	Readout and hardware in the notebook
69	**C**	DD and Markovian noise	79	**A**	ZNE extrapolation values
70	**A**	3-qubit bit-flip code	80	**C**	Interpreting ZNE

Topic-wise map of the questions

Every question appears under at least one theme; several appear under two.

Theme	Question numbers
Probability, statistics and sampling	7, 8, 13, 30, 31, 51, 52, 59
Density matrices, purity and the Bloch sphere	1, 2, 3, 4, 12, 15, 17, 18, 19
Linear operators, partial trace and fidelity	10, 11, 14, 16, 20, 40, 61
Phases, decoherence and dephasing	6, 22, 25, 34, 38, 68, 69
Pauli channels: bit, phase, bit-phase flip, depolarizing	21, 23, 33, 35, 36, 41, 44, 45
Amplitude and phase damping, T_1, T_2	5, 24, 25, 26, 27, 28, 29, 37, 39, 40
Kraus operators and CPTP maps	5, 16, 35, 43, 44, 45, 57
Measurement and readout errors	30, 31, 66, 67, 78
Coherent vs stochastic errors	9, 32, 72, 73, 76
Noise models and simulator targets in CUDA-Q	41, 42, 48, 49, 53, 54, 55, 58, 74
`Noise_Simulation.ipynb`: code and recorded results	46, 47, 50, 56, 59, 60, 74, 78
Fidelity, error rates and benchmarking	51, 56, 59, 61, 62
Error mitigation: ZNE, readout mitigation	63, 64, 65, 66, 67, 77, 79, 80
Dynamical decoupling and quantum error correction	68, 69, 70, 71, 75

Quick-marking grid

1 **C**	2 **C**	3 **B**	4 **D**	5 **A**	6 **C**	7 **A**	8 **B**	9 **A**	10 **B**
11 **C**	12 **B**	13 **A**	14 **C**	15 **D**	16 **D**	17 **A**	18 **D**	19 **D**	20 **B**
21 **D**	22 **D**	23 **C**	24 **A**	25 **C**	26 **D**	27 **D**	28 **B**	29 **A**	30 **C**
31 **A**	32 **B**	33 **B**	34 **A**	35 **B**	36 **D**	37 **B**	38 **C**	39 **A**	40 **C**
41 **B**	42 **A**	43 **D**	44 **B**	45 **C**	46 **C**	47 **A**	48 **A**	49 **C**	50 **B**
51 **D**	52 **C**	53 **D**	54 **D**	55 **A**	56 **C**	57 **D**	58 **A**	59 **B**	60 **B**
61 **A**	62 **D**	63 **D**	64 **C**	65 **C**	66 **B**	67 **D**	68 **D**	69 **C**	70 **A**
71 **B**	72 **A**	73 **B**	74 **D**	75 **A**	76 **B**	77 **C**	78 **B**	79 **A**	80 **C**

Answer Bank — questions with detailed solutions

SECTION A · Mathematical and Quantum Foundations Q.1–20

Q.1 SINGLE-CORRECT MCQ · Density matrix of a pure state

The density matrix of the pure state $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle)$ is

(A) $\frac{1}{2}\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$ **(B)** $\frac{1}{2}\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}$ **(C)** ✓ $\frac{1}{2}\begin{pmatrix}1 & -i\\ i & 1\end{pmatrix}$ **(D)** $\frac{1}{2}\begin{pmatrix}1 & i\\ -i & 1\end{pmatrix}$

Answer: (C) Detailed solution

Correct answer. (C) $\frac{1}{2}\begin{pmatrix}1 & -i\\ i & 1\end{pmatrix}$

Concepts required. A pure state's density matrix is the outer product $|\psi\rangle\langle\psi|$; the bra is the conjugate transpose of the ket.

Relevant formulae. $\rho_{ij} = \psi_i\psi_j^*$; for $\alpha\,|0\rangle + \beta\,|1\rangle$: $\rho = \begin{pmatrix}|\alpha|^2 & \alpha\beta^*\\ \alpha^*\beta & |\beta|^2\end{pmatrix}$.

Step-by-step explanation. $\rho = |\psi\rangle\langle\psi|$ with $|\psi\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ i\end{pmatrix}$ and $\langle\psi| = \frac{1}{\sqrt{2}}(1,\ -i)$ (complex conjugate transpose). Hence $\rho = \frac{1}{2}\begin{pmatrix}1\cdot 1 & 1\cdot(-i)\\ i\cdot 1 & i\cdot(-i)\end{pmatrix} = \frac{1}{2}\begin{pmatrix}1 & -i\\ i & 1\end{pmatrix}$. *Check*: Hermitian, trace 1, $\rho^2 = \rho$ (pure). Its Bloch vector is $(0, 1, 0)$.

Why the other options are incorrect.

(A) $I/2$ is the maximally mixed state; a pure state has $\rho^2 = \rho$ and non-zero coherence here.

(B) This is $|+\rangle\langle+|$; the relative phase i must appear in the off-diagonal elements.

(D) This forgets the complex conjugate in the bra: it is the density matrix of $\frac{1}{\sqrt{2}}(|0\rangle - i\,|1\rangle)$.

Key takeaway. $\rho_{01} = \alpha\beta^*$: phases live in the off-diagonal elements.

Q.2 SINGLE-CORRECT MCQ · Purity $\mathrm{Tr}\,\rho^2$

The purity $\mathrm{Tr}\,\rho^2$ of $\rho = \begin{pmatrix}0.7 & 0.3\\ 0.3 & 0.3\end{pmatrix}$ is

(A) 0.58 **(B)** 1 **(C)** ✓ 0.76 **(D)** 0.52

Answer: (C) Detailed solution

Correct answer. (C) 0.76

Concepts required. Purity $\mathrm{Tr}\,\rho^2$ distinguishes pure (1) from mixed (< 1) states; it equals the sum of squared moduli of all entries.

Relevant formulae. $\mathrm{Tr}\,\rho^2 = \sum_{ij}|\rho_{ij}|^2 = \frac{1+|\vec{r}|^2}{2}$ for a qubit.

Step-by-step explanation. $\mathrm{Tr}\,\rho^2 = \sum_{ij}|\rho_{ij}|^2 = 0.7^2+0.3^2+0.3^2+0.3^2 = 0.49+0.09+0.09+0.09 = 0.76$. *Check* with the Bloch vector: $x = 2\,\mathrm{Re}\,\rho_{01} = 0.6$, $z = \rho_{00} - \rho_{11} = 0.4$, $|\vec{r}|^2 = 0.52$ and $\mathrm{Tr}\,\rho^2 = \frac{1+|\vec{r}|^2}{2} = 0.76$. Since $0.76 < 1$ the state is mixed.

Why the other options are incorrect.

(A) 0.58 keeps only the diagonal $(0.49 + 0.09)$ and forgets both off-diagonal terms.

(B) 1 would mean a pure state, but $\det\rho = 0.12 \neq 0$.

(D) 0.52 is $|\vec{r}|^2$, not the purity $\frac{1+|\vec{r}|^2}{2}$.

Key takeaway. Purity = sum of $|\rho_{ij}|^2$; the off-diagonals count twice.

Q.3 SINGLE-CORRECT MCQ · Bloch vector from ρ

The Bloch vector (r_x, r_y, r_z) of $\rho = \begin{pmatrix} 0.8 & 0.2 - 0.1i \\ 0.2 + 0.1i & 0.2 \end{pmatrix}$ is

(A) $(0.2,\ 0.1,\ 0.6)$ **(B)** ✓ $(0.4,\ 0.2,\ 0.6)$

(C) $(0.4,\ -0.2,\ 0.6)$ **(D)** $(0.4,\ 0.2,\ 0.8)$

Answer: (B) Detailed solution

Correct answer. **(B)** $(0.4,\ 0.2,\ 0.6)$

Concepts required. Every qubit density matrix is $\frac{1}{2}(I + \vec{r}\cdot\vec{\sigma})$; the Bloch vector components are Pauli expectation values.

Relevant formulae. $r_x = 2\,\mathrm{Re}\,\rho_{10}$, $r_y = 2\,\mathrm{Im}\,\rho_{10}$, $r_z = \rho_{00} - \rho_{11}$; $\rho_{01} = \frac{r_x - i r_y}{2}$.

Step-by-step explanation. Write $\rho = \frac{1}{2}(I + r_x X + r_y Y + r_z Z) = \frac{1}{2}\begin{pmatrix} 1 + r_z & r_x - i r_y \\ r_x + i r_y & 1 - r_z \end{pmatrix}$. Then $r_z = \rho_{00} - \rho_{11} = 0.6$, and $\rho_{10} = \frac{1}{2}(r_x + i r_y) = 0.2 + 0.1i$ gives $r_x = 0.4$, $r_y = 0.2$. Equivalently $r_k = \mathrm{Tr}(\rho\sigma_k)$. $|\vec{r}|^2 = 0.56 < 1$: a valid mixed state.

Why the other options are incorrect.

(A) This forgets the factor 2 between off-diagonal entries and Bloch components.

(C) The sign of r_y is read from ρ_{01} instead of ρ_{10}: $\rho_{01} = \frac{r_x - i r_y}{2}$.

(D) $r_z = 0.8$ uses ρ_{00} instead of $\rho_{00} - \rho_{11}$.

Key takeaway. $\vec{r} = (\mathrm{Tr}\,\rho X, \mathrm{Tr}\,\rho Y, \mathrm{Tr}\,\rho Z)$.

Q.4 SINGLE-CORRECT MCQ · Validity of density matrices

Which of the following matrices is *not* a valid density matrix?

(A) $\begin{pmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{pmatrix}$ **(B)** $\begin{pmatrix} 0.5 & -0.3i \\ 0.3i & 0.5 \end{pmatrix}$ **(C)** $\begin{pmatrix} 0.9 & 0 \\ 0 & 0.1 \end{pmatrix}$ **(D)** ✓ $\begin{pmatrix} 0.6 & 0.5 \\ 0.5 & 0.4 \end{pmatrix}$

Answer: (D) Detailed solution

Correct answer. **(D)** $\begin{pmatrix} 0.6 & 0.5 \\ 0.5 & 0.4 \end{pmatrix}$

Concepts required. Valid density matrices: Hermitian, unit trace, positive semi-definite (non-negative eigenvalues).

Relevant formulae. For 2×2 Hermitian matrices with trace 1: PSD $\Leftrightarrow \det\rho \geq 0 \Leftrightarrow |\vec{r}| \leq 1$.

Step-by-step explanation. A density matrix must be Hermitian, have trace 1 and be positive semi-definite. All four are Hermitian with trace 1; for a 2×2 matrix PSD $\Leftrightarrow \det\rho \geq 0$. $\det\left(\begin{smallmatrix} 0.6 & 0.5 \\ 0.5 & 0.4 \end{smallmatrix}\right) = 0.24 - 0.25 = -0.01 < 0$: one eigenvalue is negative (a "probability" below zero). The others have determinants 0 (pure $|+\rangle$), 0.09 and $0.25 - 0.09 = 0.16$. Equivalently the Bloch vector of the first has length $\sqrt{1^2 + 0.2^2} > 1$.

Why the other options are incorrect.

(A) $|+\rangle\langle+|$: $\det = 0$, eigenvalues 1 and 0, a valid pure state.

(B) $\det = 0.25 - 0.09 = 0.16 > 0$; eigenvalues 0.8 and 0.2: valid.

(C) A diagonal matrix with entries $0.9, 0.1$ is a valid mixed state.

Key takeaway. Check the determinant (or $|\vec{r}| \leq 1$), not just trace and Hermiticity.

Q.5 SINGLE-CORRECT MCQ · Trace preservation of Kraus sets

The operators $K_0 = \begin{pmatrix} 1 & 0 \\ 0 & 0.8 \end{pmatrix}$ and $K_1 = \begin{pmatrix} 0 & b \\ 0 & 0 \end{pmatrix}$ $(b \geq 0)$ form a trace-preserving set of Kraus operators exactly when b equals

(A) ✓ 0.6 **(B)** 0.36 **(C)** 0.8 **(D)** 0.4

Answer: (A) Detailed solution

Correct answer. **(A)** 0.6

Concepts required. Kraus operators of a channel must satisfy the completeness (trace-preservation) relation.

Relevant formulae. $\sum_k K_k^\dagger K_k = I$; amplitude damping $K_0 = \mathrm{diag}(1, \sqrt{1-\gamma})$, $K_1 = \sqrt{\gamma}\,|0\rangle\langle 1|$.

Step-by-step explanation. Trace preservation requires $\sum_k K_k^\dagger K_k = I$. $K_0^\dagger K_0 = \mathrm{diag}(1, 0.64)$ and $K_1^\dagger K_1 = \left(\begin{smallmatrix} 0 & 0 \\ b & 0 \end{smallmatrix}\right)\left(\begin{smallmatrix} 0 & b \\ 0 & 0 \end{smallmatrix}\right) = \mathrm{diag}(0, b^2)$. So $0.64 + b^2 = 1$, $b = 0.6$. This is amplitude damping with $\gamma = 0.36$: $K_0 = \mathrm{diag}(1, \sqrt{1-\gamma})$, $K_1 = \sqrt{\gamma}\,|0\rangle\langle 1|$.

Why the other options are incorrect.

(B) $b^2 = 0.36$ is required, so $b = 0.6$; choosing $b = 0.36$ confuses b with b^2 and gives $0.64 + 0.13 = 0.77 \neq 1$.

(C) $b = 0.8$ gives $K_0^\dagger K_0 + K_1^\dagger K_1 = \mathrm{diag}(1, 0.64{+}0.64) = \mathrm{diag}(1, 1.28)$: probability would be created.

(D) $b = 0.4$ gives $\mathrm{diag}(1, 0.64 + 0.16) = \mathrm{diag}(1, 0.80)$: 20 % of the $|1\rangle$ population would disappear.

Key takeaway. Completeness: $K_0^\dagger K_0 + K_1^\dagger K_1 = I$ fixes $b^2 = 1 - 0.64$.

Q.6 SINGLE-CORRECT MCQ · Random phases → dephasing

A qubit is prepared in $\frac{1}{\sqrt{2}}(|0\rangle + e^{i\varphi}\,|1\rangle)$ where the phase φ is random and uniformly distributed in $[-\frac{\pi}{2}, \frac{\pi}{2}]$. The average value of $\langle X \rangle$ is

(A) $\pi/4 \approx 0.79$ **(B)** $1/2$ **(C)** ✓ $2/\pi \approx 0.64$ **(D)** 0

Answer: (C) Detailed solution

Correct answer. **(C)** $2/\pi \approx 0.64$

Concepts required. Averaging over a random phase reduces coherence: the origin of dephasing.

Relevant formulae. $\langle X \rangle = \cos\varphi$; $\mathbb{E}\cos\varphi = \frac{1}{2\Delta}\int_{-\Delta}^{\Delta} \cos\varphi\, d\varphi = \frac{\sin\Delta}{\Delta}$.

Step-by-step explanation. For fixed φ, $\langle X \rangle = 2\,\mathrm{Re}(\alpha^*\beta) = \cos\varphi$. Averaging over the uniform distribution: $\frac{1}{\pi}\int_{-\pi/2}^{\pi/2} \cos\varphi\, d\varphi = \frac{1}{\pi}[\sin\varphi]_{-\pi/2}^{\pi/2} = \frac{2}{\pi} \approx 0.637$. The averaged state $\bar{\rho}$ has its off-diagonal element reduced from $\frac{1}{2}$ to $\frac{1}{\pi}$: random phases are the origin of *dephasing*. Populations ($\langle Z \rangle = 0$) are unchanged.

Why the other options are incorrect.

(A) $\frac{\pi}{4}$ is the inverse of the correct ratio ($\frac{\Delta}{\sin\Delta}$ instead of $\frac{\sin\Delta}{\Delta}$ with $\Delta = \frac{\pi}{2}$) and exceeds the correct value.

(B) $\frac{1}{2}$ is not the mean of $\cos\varphi$ on $[-\frac{\pi}{2}, \frac{\pi}{2}]$; it would be the mean of $\cos^2\varphi$ (the average of $P(+)$-like quantities), not of $\langle X \rangle$.

(D) 0 would require the phase to be uniform over the whole circle $[-\pi, \pi]$, where positive and negative $\cos\varphi$ cancel; here $\cos\varphi \geq 0$ throughout.

Key takeaway. Random phases shrink $\langle X \rangle$, not the populations.

Q.7 SINGLE-CORRECT MCQ · Error-free probability $(1-p)^n$

A circuit has 20 gates; each fails independently with probability $p = 0.01$. The probability that *at least one* gate fails is

(A) ✓ 0.182 **(B)** 0.818 **(C)** 0.200 **(D)** 0.165

Answer: (A) Detailed solution

Correct answer. **(A)** 0.182
Concepts required. Independent failures; complement rule.
Relevant formulae. $P(\geq 1) = 1 - (1-p)^n$; $P(\text{exactly } k) = \binom{n}{k} p^k (1-p)^{n-k}$.
Step-by-step explanation. $P(\text{no failure}) = (1-p)^{20} = 0.99^{20} = e^{20 \ln 0.99} = e^{-0.2010} = 0.818$. Hence $P(\geq 1) = 1 - 0.818 = 0.182$. The linear estimate $20p = 0.200$ over-counts runs with two or more failures; $P(\text{exactly one}) = 20(0.01)(0.99)^{19} = 0.165$.
Why the other options are incorrect.
(B) 0.818 is the probability of *no* failure.
(C) $np = 0.2$ is a first-order approximation that over-counts.
(D) 0.165 is the probability of exactly one failure.
Key takeaway. Error-free probability $(1-p)^G$ — the notebook's lower bound.

Q.8 SINGLE-CORRECT MCQ · Shot-noise standard error

An outcome has true probability 0.9. It is estimated from 10 000 shots. The standard error of the estimate is

(A) 0.03 **(B)** ✓ 0.003 **(C)** 0.0009 **(D)** 0.009

Answer: (B) Detailed solution

Correct answer. **(B)** 0.003
Concepts required. Binomial sampling error of an estimated probability.
Relevant formulae. $\text{SE} = \sqrt{P(1-P)/S}$.
Step-by-step explanation. The count is binomial; the estimated frequency has $\text{SE} = \sqrt{P(1-P)/S} = \sqrt{0.9 \cdot 0.1/10\,000} = \sqrt{9 \times 10^{-6}} = 0.003$. Shot noise is a *statistical* error of the estimate, not a property of the quantum state: it shrinks like $1/\sqrt{S}$ even on a noiseless simulator.
Why the other options are incorrect.
(A) $0.03 = \sqrt{0.09/100}$ is the standard error for only 100 shots; with 10^4 shots it is ten times smaller.
(C) $0.0009 = \frac{0.09}{100}$ is neither the variance 9×10^{-6} nor its square root 0.003.
(D) 0.009 omits the square root: $\frac{0.9 \times 0.1}{10} = 0.009$ mixes the variance with a wrong divisor.
Key takeaway. Shot noise $\propto 1/\sqrt{S}$: 10 000 shots give ± 0.003.

Q.9 SINGLE-CORRECT MCQ · Coherent angle error

An X gate is implemented as $R_X(\pi+\varepsilon)$ instead of $R_X(\pi)$, with $\varepsilon = 0.1$ rad. Applied to $|0\rangle$, the probability of measuring 0 (an error) is closest to

(A) ✓ 0.0025 **(B)** 0.01 **(C)** 0.1 **(D)** 0.05

Answer: (A) Detailed solution

Correct answer. **(A)** 0.0025
Concepts required. Rotation gates and coherent angle errors.
Relevant formulae. $R_X(\theta) = \cos\frac{\theta}{2} I - i\sin\frac{\theta}{2} X$; $P(0) = \cos^2\frac{\pi+\varepsilon}{2} = \sin^2\frac{\varepsilon}{2} \approx \frac{\varepsilon^2}{4}$.
Step-by-step explanation. $R_X(\theta)|0\rangle = \cos\frac{\theta}{2}|0\rangle - i\sin\frac{\theta}{2}|1\rangle$, so $P(0) = \cos^2\frac{\pi+\varepsilon}{2} = \sin^2\frac{\varepsilon}{2}$. For $\varepsilon = 0.1$: $\sin^2(0.05) = 0.0025$. A small *coherent* angle error gives an error probability quadratic in ε (but such errors add up in amplitude when repeated — see Section D).
Why the other options are incorrect.
(B) $0.01 = \varepsilon^2$ forgets the half angle.
(C) $0.1 = \varepsilon$ treats the angle as a probability.
(D) $0.05 = \varepsilon/2$ is an amplitude, not a probability.
Key takeaway. A coherent error ε costs $\approx \varepsilon^2/4$ in probability per gate.

Q.10 SINGLE-CORRECT MCQ · Fidelity with a mixed state

The fidelity $F = \langle+|\rho|+\rangle$ between the target $|+\rangle$ and the mixed state $\rho = 0.8\,|+\rangle\langle+| + 0.2\,\frac{I}{2}$ is

(A) 0.6 **(B)** ✓ 0.9 **(C)** 0.8 **(D)** 0.95

Answer: (B) Detailed solution

Correct answer. **(B)** 0.9
Concepts required. Fidelity of a mixed state with a pure target is a linear function of ρ.
Relevant formulae. $F = \langle\psi|\rho|\psi\rangle$; $\langle\psi|\frac{I}{2}|\psi\rangle = \frac{1}{2}$.
Step-by-step explanation. By linearity, $\langle+|\rho|+\rangle = 0.8\,\langle+|+\rangle\,\langle+|+\rangle + 0.2\cdot\frac{1}{2}\,\langle+|+\rangle = 0.8 + 0.1 = 0.9$. The maximally mixed part still overlaps the target with probability $\frac{1}{2}$. (For a pure target, $F = \langle\psi|\rho|\psi\rangle$; some texts use the square root.)
Why the other options are incorrect.
(A) 0.6 subtracts the noisy part instead of adding its overlap.
(C) 0.8 ignores that the mixed part still overlaps the target.
(D) $0.95 \approx \sqrt{0.9}$ is the root fidelity convention.
Key takeaway. $F = (1-q) + q/d$ for a target mixed with white noise.

Q.11 SINGLE-CORRECT MCQ · Partial trace; entanglement and mixedness

For the two-qubit state $|\psi\rangle = \sqrt{0.8}\,|00\rangle + \sqrt{0.2}\,|11\rangle$, the purity $\mathrm{Tr}\,\rho_A^2$ of the reduced state of qubit A is

(A) 0.8 **(B)** 0.5 **(C)** ✓ 0.68 **(D)** 1

Answer: (C) Detailed solution

Correct answer. **(C)** 0.68
Concepts required. Partial trace; entanglement makes subsystems mixed.
Relevant formulae. $\rho_A = \mathrm{Tr}_B\,|\psi\rangle\langle\psi|$; for $\sum_i \sqrt{\lambda_i}\,|ii\rangle$: $\rho_A = \mathrm{diag}(\lambda_i)$.
Step-by-step explanation. $\rho_A = \mathrm{Tr}_B\,|\psi\rangle\langle\psi| = 0.8\,|0\rangle\langle0| + 0.2\,|1\rangle\langle1|$ (the cross terms $|00\rangle\langle11|$ have orthogonal B parts and vanish). $\mathrm{Tr}\,\rho_A^2 = 0.64 + 0.04 = 0.68$. The global state is pure, but its parts

are mixed because they are entangled — the same mechanism by which a qubit entangled with its environment decoheres.

Why the other options are incorrect.

(A) 0.8 is the population $\rho_{A,00}$, not the purity $\mathrm{Tr}\,\rho_A^2 = 0.8^2 + 0.2^2$.

(B) $\frac{1}{2}$ is the purity for the maximally entangled state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, where $\rho_A = \frac{I}{2}$; here the weights are 0.8 and 0.2.

(D) 1 would hold for a product state such as $|00\rangle$; entanglement with qubit B makes ρ_A mixed even though the global state is pure.

Key takeaway. Decoherence = entanglement with an environment we trace out.

Q.12 SINGLE-CORRECT MCQ · Eigenvalues of a qubit ρ

The larger eigenvalue of $\rho = \begin{pmatrix} 0.7 & 0.3 \\ 0.3 & 0.3 \end{pmatrix}$ is closest to

(A) 0.700 **(B)** ✓ 0.861 **(C)** 1.000 **(D)** 0.760

Answer: (B) Detailed solution

Correct answer. **(B)** 0.861

Concepts required. Eigenvalues of a qubit density matrix from its Bloch length.

Relevant formulae. $\lambda_\pm = \frac{1 \pm |\vec{r}|}{2}$; $\lambda_+\lambda_- = \det\rho$.

Step-by-step explanation. For a qubit, $\lambda_\pm = \frac{1 \pm |\vec{r}|}{2}$. Here $\vec{r} = (0.6, 0, 0.4)$, $|\vec{r}| = \sqrt{0.52} = 0.7211$, so $\lambda_+ = 0.8606$, $\lambda_- = 0.1394$. *Check*: $\lambda_+ + \lambda_- = 1 = \mathrm{Tr}\,\rho$ and $\lambda_+\lambda_- = 0.12 = \det\rho = 0.21 - 0.09$; $\lambda_+^2 + \lambda_-^2 = 0.76 = \mathrm{Tr}\,\rho^2$.

Why the other options are incorrect.

(A) 0.700 is the diagonal entry ρ_{00}; diagonal entries equal eigenvalues only when the off-diagonal elements vanish, and here $\rho_{01} = 0.3$.

(C) An eigenvalue 1 would make the other eigenvalue 0 and the state pure, but $\det\rho = 0.12 \neq 0$.

(D) 0.760 is the purity $\lambda_+^2 + \lambda_-^2$, not an eigenvalue.

Key takeaway. Diagonal entries are not eigenvalues unless the coherence is zero.

Q.13 SINGLE-CORRECT MCQ · Independent qubits: tensor product

Two independent qubits are in the states $\rho_A = \mathrm{diag}(0.9, 0.1)$ and $\rho_B = \mathrm{diag}(0.8, 0.2)$. In the state $\rho_A \otimes \rho_B$, the probability that both qubits give the *same* outcome is

(A) ✓ 0.74 **(B)** 0.50 **(C)** 0.26 **(D)** 0.72

Answer: (A) Detailed solution

Correct answer. **(A)** 0.74

Concepts required. Product states of independent qubits; tensor products of diagonal matrices.

Relevant formulae. $P(ab) = P_A(a)P_B(b)$.

Step-by-step explanation. $\rho_A \otimes \rho_B = \mathrm{diag}(0.9 \cdot 0.8,\ 0.9 \cdot 0.2,\ 0.1 \cdot 0.8,\ 0.1 \cdot 0.2) = \mathrm{diag}(0.72, 0.18, 0.08, 0.02)$ in the order $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. Same outcome: $0.72 + 0.02 = 0.74$.

Why the other options are incorrect.

(B) 0.5 would hold for unbiased qubits; both qubits are strongly biased towards 0.

(C) $0.26 = 0.18 + 0.08$ is the probability of *different* outcomes (01 or 10), the complement of the answer.

(D) 0.72 counts only the outcome 00; the outcome 11 (probability $0.1 \times 0.2 = 0.02$) also has both qubits equal.

Key takeaway. Independent noise multiplies probabilities.

Q.14 Single-correct MCQ · Conjugation $X\rho X$ on the Bloch sphere

A qubit with Bloch vector (r_x, r_y, r_z) is acted on by $\rho \mapsto X\rho X$. Its new Bloch vector is

(A) $(-r_x,\ -r_y,\ -r_z)$ **(B)** $(-r_x,\ r_y,\ r_z)$ **(C)** ✓ $(r_x,\ -r_y,\ -r_z)$ **(D)** $(r_x,\ r_y,\ -r_z)$

Answer: (C) Detailed solution

Correct answer. **(C)** $(r_x,\ -r_y,\ -r_z)$
Concepts required. Conjugation by a Pauli is a π rotation of the Bloch sphere.
Relevant formulae. $XXX = X$, $XYX = -Y$, $XZX = -Z$.
Step-by-step explanation. $X\rho X = \frac{1}{2}(I + r_x XXX + r_y XYX + r_z XZX)$. Since X commutes with itself and anticommutes with Y and Z: $XXX = X$, $XYX = -Y$, $XZX = -Z$. So $(r_x, r_y, r_z) \mapsto (r_x, -r_y, -r_z)$: a rotation by π about the x-axis. The bit-flip channel $(1-p)\rho + pX\rho X$ therefore keeps r_x and multiplies r_y, r_z by $1 - 2p$.
Why the other options are incorrect.
(A) Inverting all three components is not a rotation (not a unitary map).
(B) This is conjugation by Z or by a rotation about another axis, not by X.
(D) Only r_z flipped would be a reflection, not a unitary rotation.
Key takeaway. $X\rho X$: keep r_x, flip r_y, r_z.

Q.15 Matching lists · Bloch vectors of states

Match each state (List-I) with its Bloch vector (List-II).

List-I		List-II	
(P)	$\lvert 1\rangle$	**(1)**	$(0,\ 0,\ 0.5)$
(Q)	$\frac{I}{2}$	**(2)**	$(0,\ 0,\ 0)$
(R)	$\frac{1}{2}\begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix}$	**(3)**	$(0,\ 0,\ -1)$
(S)	$\mathrm{diag}(0.75,\ 0.25)$	**(4)**	$(0,\ 1,\ 0)$

(A) P→3, Q→1, R→4, S→2 **(B)** P→3, Q→2, R→1, S→4
(C) P→1, Q→2, R→4, S→3 **(D)** ✓ P→3, Q→2, R→4, S→1

Answer: (D) Detailed solution

Correct answer. **(D)** P→3, Q→2, R→4, S→1
Concepts required. Bloch vectors of pure, mixed and maximally mixed states.
Relevant formulae. $r_z = \rho_{00} - \rho_{11}$, $r_x + ir_y = 2\rho_{10}$.
Step-by-step explanation. Use $r_z = \rho_{00} - \rho_{11}$, $r_x + ir_y = 2\rho_{10}$.

- $\lvert 1\rangle$: south pole $(0, 0, -1)$ (3).
- $I/2$: the centre of the ball (2) — maximally mixed.
- $\rho_{10} = \frac{i}{2}$: $r_y = 1$, i.e. $(0, 1, 0)$ (4), the pure state $\frac{1}{\sqrt{2}}(\lvert 0\rangle + i\lvert 1\rangle)$.
- $\mathrm{diag}(0.75, 0.25)$: $r_z = 0.5$ (1), a mixed state on the z-axis.

Hence P→3, Q→2, R→4, S→1.
Why the other options are incorrect.
(A) Q→1, S→2 confuses $I/2$ (centre) with a partially mixed state.
(B) R→1, S→4 gives a diagonal matrix a y-component, which needs an imaginary off-diagonal.

(C) P→1, S→3 swaps $|1\rangle$ and the mixed state: $|1\rangle$ is pure, length 1.
Key takeaway. Length = purity information; direction = which state.

Q.16 MATCHING LISTS · Hermitian, unitary, projector, Kraus

Match each matrix (List-I) with its property (List-II).

List-I		List-II	
(P)	$\left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)$	**(1)**	unitary but not Hermitian
(Q)	$\frac{1}{2}\left(\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right)$	**(2)**	Hermitian and unitary
(R)	$\left(\begin{smallmatrix}1&0\\0&i\end{smallmatrix}\right)$	**(3)**	neither Hermitian nor unitary (a Kraus operator of amplitude damping, up to a factor)
(S)	$\left(\begin{smallmatrix}0&1\\0&0\end{smallmatrix}\right)$	**(4)**	a rank-1 projector and a valid density matrix

(A) P→4, Q→2, R→1, S→3
(B) P→2, Q→4, R→3, S→1
(C) P→1, Q→4, R→2, S→3
(D) ✓ P→2, Q→4, R→1, S→3

Answer: (D) Detailed solution

Correct answer. **(D)** P→2, Q→4, R→1, S→3
Concepts required. Hermitian, unitary, projector and density-matrix properties.
Relevant formulae. Hermitian $A = A^\dagger$; unitary $U^\dagger U = I$; projector $P^2 = P = P^\dagger$.
Step-by-step explanation.

- X: $X^\dagger = X$ and $X^\dagger X = I$ (2); it has trace 0, so it is not a density matrix.
- $|+\rangle\langle+|$: Hermitian, $P^2 = P$, trace 1, PSD (4); not unitary (det = 0).
- $S = \mathrm{diag}(1, i)$: $S^\dagger S = I$ but $S^\dagger = \mathrm{diag}(1, -i) \neq S$ (1).
- $|0\rangle\langle 1|$: nilpotent, $\neq$ its adjoint, not invertible (3); $\sqrt{\gamma}\,|0\rangle\langle 1|$ is the "jump" of amplitude damping.

Hence P→2, Q→4, R→1, S→3.
Why the other options are incorrect.
(A) P→4 makes X a density matrix, but $\mathrm{Tr}\, X = 0$ and $X^2 = I \neq X$ (not a projector).
(B) R→3 calls S non-unitary, but $S^\dagger S = \mathrm{diag}(1, -i \cdot i) = I$.
(C) P→1 calls X non-Hermitian, but $X^\dagger = X$; and R→2 calls S Hermitian, but $S^\dagger = \mathrm{diag}(1, -i) \neq S$.
Key takeaway. Kraus operators need not be Hermitian or unitary; only $\sum K^\dagger K = I$.

Q.17 Assertion–Reason · Mixed states inside the Bloch ball

Assertion (A): Every mixed single-qubit state lies strictly inside the Bloch ball ($|\vec{r}| < 1$).

Reason (R): For $\rho = \frac{1}{2}(I + \vec{r}\cdot\vec{\sigma})$ one has $\mathrm{Tr}\,\rho^2 = \frac{1+|\vec{r}|^2}{2}$, which equals 1 only when $|\vec{r}| = 1$.

(A) ✓ Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Answer: (A) Detailed solution

Correct answer. **(A)** Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

Concepts required. Purity and Bloch length.

Relevant formulae. $(\vec{r}\cdot\vec{\sigma})^2 = |\vec{r}|^2 I$; $\mathrm{Tr}\,\rho^2 = \frac{1+|\vec{r}|^2}{2}$.

Step-by-step explanation. **R is true**: $\rho^2 = \frac{1}{4}(I + 2\vec{r}\cdot\vec{\sigma} + (\vec{r}\cdot\vec{\sigma})^2) = \frac{1}{4}((1+|\vec{r}|^2)I + 2\vec{r}\cdot\vec{\sigma})$ because $(\vec{r}\cdot\vec{\sigma})^2 = |\vec{r}|^2 I$; the trace is $\frac{1+|\vec{r}|^2}{2}$. **A is true**: "mixed" means $\mathrm{Tr}\,\rho^2 < 1$, which by R is exactly $|\vec{r}| < 1$. R is the reason for A. Answer: (A).

Why the other options are incorrect.

(B) requires that R does not explain A; but A follows directly from R, since "mixed" means $\mathrm{Tr}\,\rho^2 < 1$.

(C) requires R to be false, but $\mathrm{Tr}\,\rho^2 = \frac{1+|\vec{r}|^2}{2}$ follows from $(\vec{r}\cdot\vec{\sigma})^2 = |\vec{r}|^2 I$ and $\mathrm{Tr}\,\sigma_k = 0$.

(D) requires A to be false, but no mixed state can reach the surface: $|\vec{r}| = 1$ forces $\mathrm{Tr}\,\rho^2 = 1$.

Key takeaway. Mixed ⇔ inside the Bloch ball; noise shrinks Bloch vectors.

Q.18 Statement I / II · Density-matrix conditions; Born rule

Statement I: Every 2×2 Hermitian matrix with trace 1 is a valid density matrix.

Statement II: The diagonal entries of a density matrix in the computational basis are the probabilities of the outcomes of a computational-basis measurement.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) ✓ Statement I is false, but Statement II is true

Answer: (D) Detailed solution

Correct answer. **(D)** Statement I is false, but Statement II is true

Concepts required. Density-matrix conditions; Born rule for mixed states.

Relevant formulae. $P(k) = \mathrm{Tr}(\Pi_k \rho) = \rho_{kk}$.

Step-by-step explanation. **I is false**: positivity is also required; $\mathrm{diag}(1.2, -0.2)$ is Hermitian with trace 1 but has a negative eigenvalue. **II is true**: $P(k) = \mathrm{Tr}(|k\rangle\langle k|\,\rho) = \rho_{kk}$ (Born rule for mixed states). This is how the notebook turns an exact noisy ρ into outcome probabilities on the density-matrix path. Answer: (D).

Why the other options are incorrect.

(A) requires Statement I to be true, but I omits positivity: $\mathrm{diag}(1.2, -0.2)$ is Hermitian with trace 1 and is not a state.

(B) requires Statement II to be false, but $\rho_{kk} = \mathrm{Tr}(|k\rangle\langle k|\,\rho)$ is exactly the Born-rule probability of outcome k.

(C) asserts I true and II false — exactly the reverse of the correct judgement.

Key takeaway. Probabilities are on the diagonal; positivity is not automatic.

Paragraph for Questions 19 and 20

A qubit should have been prepared in $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. State tomography of the real (noisy) preparation gives the density matrix

$$\rho = \begin{pmatrix} 0.55 & 0.35 \\ 0.35 & 0.45 \end{pmatrix}.$$

Q.19 PARAGRAPH BASED · Tomography: Bloch vector, purity

The Bloch vector and the purity of the prepared state are

(A) $(0.7,\ 0,\ 0.1)$ and 0.50 **(B)** $(0.7,\ 0,\ 0.55)$ and 0.85

(C) $(0.35,\ 0,\ 0.1)$ and 0.75 **(D)** ✓ $(0.7,\ 0,\ 0.1)$ and 0.75

Answer: (D) Detailed solution

Correct answer. (D) $(0.7,\ 0,\ 0.1)$ and 0.75
Concepts required. Bloch vector and purity of a measured density matrix.
Relevant formulae. $r_x = 2\,\mathrm{Re}\,\rho_{01}$, $r_z = \rho_{00} - \rho_{11}$; $\mathrm{Tr}\,\rho^2 = \frac{1+|\vec{r}|^2}{2}$.
Step-by-step explanation. $r_x = 2\,\mathrm{Re}\,\rho_{01} = 0.7$, $r_y = 0$ (real matrix), $r_z = 0.55 - 0.45 = 0.1$. $\mathrm{Tr}\,\rho^2 = 0.55^2 + 0.45^2 + 2(0.35)^2 = 0.3025 + 0.2025 + 0.245 = 0.75 = \frac{1+0.49+0.01}{2}$. The vector is shorter than 1 (mixed: decoherence) and tilted towards $|0\rangle$ (a population imbalance, e.g. relaxation).
Why the other options are incorrect.

(A) Purity 0.50 belongs to the maximally mixed state $I/2$; here the Bloch vector has length $\sqrt{0.5} \approx 0.71$, so the purity is 0.75.

(B) $r_z = 0.55$ uses ρ_{00} instead of $\rho_{00} - \rho_{11}$, and 0.85 is the fidelity with $|+\rangle$, not the purity.

(C) $r_x = 0.35$ forgets the factor 2 in $r_x = 2\,\mathrm{Re}\,\rho_{01}$ (and would give a different purity).

Key takeaway. Read $\vec{r}$ off ρ, then purity from $|\vec{r}|$.

Q.20 PARAGRAPH BASED · Tomography: fidelity, probabilities

The fidelity $\langle +|\rho|+\rangle$ of the preparation, and the probability that a computational-basis measurement gives 1, are

(A) 0.70 and 0.45 **(B)** ✓ 0.85 and 0.45 **(C)** 0.92 and 0.45 **(D)** 0.85 and 0.55

Answer: (B) Detailed solution

Correct answer. (B) 0.85 and 0.45
Concepts required. Fidelity with $|+\rangle$ and computational-basis probabilities.
Relevant formulae. $\langle +|\rho|+\rangle = \frac{1+r_x}{2}$; $P(1) = \rho_{11}$.
Step-by-step explanation. $\langle +|\rho|+\rangle = \frac{1}{2}(\rho_{00} + \rho_{01} + \rho_{10} + \rho_{11}) = \frac{1}{2}(0.55 + 0.35 + 0.35 + 0.45) = 0.85 = \frac{1+r_x}{2}$. $P(1) = \rho_{11} = 0.45$. ($0.92 = \sqrt{0.85}$ is the root fidelity used in some texts; 0.70 is r_x itself.)
Why the other options are incorrect.

(A) 0.70 is the Bloch component r_x; the fidelity is $\frac{1+r_x}{2} = 0.85$.

(C) $0.92 = \sqrt{0.85}$ is the "root fidelity" convention; the question defines $F = \langle +|\rho|+\rangle$.

(D) $0.55 = \rho_{00}$ is the probability of outcome 0; the probability of 1 is $\rho_{11} = 0.45$.

Key takeaway. $F(|+\rangle, \rho) = \frac{1+r_x}{2}$.

SECTION B · Quantum Noise Fundamentals Q.21–40

Q.21 SINGLE-CORRECT MCQ · Fixed points of bit flip

Which single-qubit state is left *unchanged* by the bit-flip channel $\rho \mapsto (1-p)\rho + pX\rho X$ for every p?

(A) $\frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle)$ **(B)** $|1\rangle$
(C) $|0\rangle$ **(D)** ✓ $|+\rangle$

Answer: (D) Detailed solution

Correct answer. (D) $|+\rangle$
Concepts required. Fixed points of a Pauli channel: eigenstates of the error operator.
Relevant formulae. Bit flip: $(r_x, r_y, r_z) \mapsto (r_x, (1-2p)r_y, (1-2p)r_z)$.
Step-by-step explanation. A state is invariant iff $X\rho X = \rho$, i.e. its Bloch vector satisfies $(r_x, -r_y, -r_z) = (r_x, r_y, r_z)$: $r_y = r_z = 0$. $|+\rangle$ has $\vec{r} = (1,0,0)$ (indeed $X|+\rangle = |+\rangle$). $|0\rangle$, $|1\rangle$ have $r_z = \pm 1$ and become mixed; $\frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle)$ has $r_y = 1$, which shrinks to $1-2p$. The eigenstates of the error operator are immune to that error.
Why the other options are incorrect.
(A) This state has $r_y = 1$, which shrinks to $1-2p$.
(B) $|1\rangle$ has $r_z = -1$, which shrinks as well.
(C) $|0\rangle$ has $r_z = 1$, which shrinks to $1-2p$.
Key takeaway. States along the error's axis are immune to it.

Q.22 SINGLE-CORRECT MCQ · Phase flip on $|+\rangle$

A phase-flip channel with $p = 0.1$ acts on $|+\rangle$. The state is then measured in the X basis. The probability of the outcome "+" is

(A) 0.95 **(B)** 1.0 **(C)** 0.8 **(D)** ✓ 0.9

Answer: (D) Detailed solution

Correct answer. (D) 0.9
Concepts required. Phase flips act like $|+\rangle \leftrightarrow |-\rangle$.
Relevant formulae. $\rho' = (1-p)\rho + pZ\rho Z$; $\langle X\rangle \mapsto (1-2p)\langle X\rangle$; $P(+) = \frac{1+\langle X\rangle}{2}$.
Step-by-step explanation. $\rho' = 0.9\,|+\rangle\langle+| + 0.1Z\,|+\rangle\langle+|\,Z = 0.9\,|+\rangle\langle+| + 0.1\,|-\rangle\langle-|$ (since $Z|+\rangle = |-\rangle$). So $P(+) = 0.9$ and $\langle X\rangle = 0.9 - 0.1 = 1 - 2p = 0.8$. In the Z basis nothing changes ($P(0) = P(1) = \frac{1}{2}$): phase errors are invisible there.
Why the other options are incorrect.
(A) 0.95 uses $p/2$ as the error probability; a Z applied with probability p turns $|+\rangle$ into $|-\rangle$ with probability p.
(B) 1.0 would mean the phase flip had no effect; that is true only for Z-basis measurements, not in the X basis.
(C) $0.8 = 1-2p$ is the expectation value $\langle X\rangle = P(+) - P(-)$, not the probability $P(+)$.
Key takeaway. Phase errors appear only in a rotated basis.

Q.23 SINGLE-CORRECT MCQ · Depolarizing flip probability

A depolarizing channel $\rho \mapsto (1-p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z)$ with $p = 0.03$ acts on $|0\rangle$. The probability of then measuring 1 is

(A) 0.03 **(B)** 0.04 **(C)** ✓ 0.02 **(D)** 0.01

Answer: (C) Detailed solution

Correct answer. (C) 0.02
Concepts required. Depolarizing channel in CUDA-Q's convention.
Relevant formulae. $\mathcal{E}(\rho) = (1-p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z)$, $\vec{r} \mapsto (1 - \frac{4p}{3})\vec{r}$.
Step-by-step explanation. X and Y flip $|0\rangle$ to $|1\rangle$ (up to a phase), Z does not. So $P(1) = \frac{p}{3} + \frac{p}{3} = \frac{2p}{3} = 0.02$. Equivalently the Bloch vector shrinks by $1 - \frac{4p}{3} = 0.96$: $r_z = 0.96$, $P(1) = \frac{1-r_z}{2} = 0.02$. (This is CUDA-Q's `DepolarizationChannel(p)` convention; the same channel is $\rho \mapsto (1-\lambda)\rho + \lambda\frac{I}{2}$ with $\lambda = \frac{4p}{3}$.)
Why the other options are incorrect.
(A) $0.03 = p$ treats all three Paulis as flips, but $Z|0\rangle = |0\rangle$ does not change the outcome.
(B) $0.04 = \frac{4p}{3}$ is the reduction $1 - r_z$ of the Bloch component, twice the flip probability $\frac{1-r_z}{2}$.
(D) $0.01 = p/3$ counts only X; $Y = iXZ$ also maps $|0\rangle$ to $|1\rangle$ (up to a phase).
Key takeaway. Two of the three Paulis flip a Z outcome: $P = \frac{2p}{3}$.

Q.24 SINGLE-CORRECT MCQ · Amplitude damping on $|+\rangle$

Amplitude damping with $\gamma = 0.36$ acts on $|+\rangle$. The resulting density matrix is

(A) ✓ $\begin{pmatrix} 0.68 & 0.40 \\ 0.40 & 0.32 \end{pmatrix}$ **(B)** $\begin{pmatrix} 0.68 & 0.32 \\ 0.32 & 0.32 \end{pmatrix}$ **(C)** $\begin{pmatrix} 0.50 & 0.40 \\ 0.40 & 0.50 \end{pmatrix}$ **(D)** $\begin{pmatrix} 0.68 & 0.50 \\ 0.50 & 0.32 \end{pmatrix}$

Answer: (A) Detailed solution

Correct answer. (A) $\begin{pmatrix} 0.68 & 0.40 \\ 0.40 & 0.32 \end{pmatrix}$
Concepts required. Amplitude damping: energy relaxation to $|0\rangle$.
Relevant formulae. $\rho_{11} \mapsto (1-\gamma)\rho_{11}$, $\rho_{00} \mapsto \rho_{00} + \gamma\rho_{11}$, $\rho_{01} \mapsto \sqrt{1-\gamma}\,\rho_{01}$.
Step-by-step explanation. With $K_0 = \mathrm{diag}(1, \sqrt{1-\gamma})$, $K_1 = \sqrt{\gamma}\,|0\rangle\langle 1|$: $\rho'_{11} = (1-\gamma)\rho_{11}$, $\rho'_{00} = \rho_{00} + \gamma\rho_{11}$, $\rho'_{01} = \sqrt{1-\gamma}\,\rho_{01}$. For $|+\rangle$ ($\rho_{ij} = \frac{1}{2}$): $\rho'_{11} = 0.64 \cdot 0.5 = 0.32$, $\rho'_{00} = 0.5 + 0.18 = 0.68$, $\rho'_{01} = 0.8 \cdot 0.5 = 0.40$. Populations move towards $|0\rangle$ and the coherence shrinks by $\sqrt{1-\gamma}$, not by $1-\gamma$.
Why the other options are incorrect.
(B) Scaling the coherence by $1 - \gamma = 0.64$ instead of $\sqrt{1-\gamma} = 0.8$ confuses the population decay with the amplitude decay.
(C) Leaving the populations at 0.5 ignores the jump $|1\rangle \to |0\rangle$; that would be phase damping, not amplitude damping.
(D) Keeping the coherence at 0.5 ignores that the no-jump operator $\mathrm{diag}(1, \sqrt{1-\gamma})$ also shrinks ρ_{01}.
Key takeaway. T_1 decay also costs coherence, at half the rate.

Q.25 Single-correct MCQ · Phase damping ≡ phase flip

Phase damping with parameter λ multiplies the off-diagonal elements by $\sqrt{1-\lambda}$; a phase-flip channel multiplies them by $1-2p$. The phase-flip probability p that has the same effect as phase damping with $\lambda = 0.36$ is

(A) 0.18 **(B)** 0.36 **(C)** ✓ 0.10 **(D)** 0.20

Answer: (C) Detailed solution

Correct answer. (C) 0.10
Concepts required. Different Kraus sets can represent the same channel.
Relevant formulae. Phase flip: coherence $\times(1-2p)$; phase damping: coherence $\times\sqrt{1-\lambda}$.
Step-by-step explanation. Both channels leave the populations unchanged, so they are equal iff they scale the coherence by the same factor: $1-2p = \sqrt{1-0.36} = 0.8 \Rightarrow p = 0.1$. Phase damping and the phase-flip channel are the same channel in different Kraus representations (Kraus operators are not unique).
Why the other options are incorrect.
(A) $0.18 = \lambda/2$ equates $2p$ with λ, which is only the first-order approximation of $1-2p = \sqrt{1-\lambda}$.
(B) 0.36 sets $p = \lambda$; then $1-2p = 0.28 \neq 0.8$.
(D) $0.20 = 1-\sqrt{1-\lambda}$ forgets the factor 2 in the phase-flip coherence factor $1-2p$.
Key takeaway. Phase damping ≡ phase flip with $1-2p = \sqrt{1-\lambda}$.

Q.26 Single-correct MCQ · T_1 half-life

A qubit with $T_1 = 50\,\mu$s starts in $|1\rangle$. After what time is $P(1) = \frac{1}{2}$?

(A) $50\,\mu$s **(B)** $69.3\,\mu$s **(C)** $25\,\mu$s **(D)** ✓ $34.7\,\mu$s

Answer: (D) Detailed solution

Correct answer. (D) $34.7\,\mu$s
Concepts required. Exponential T_1 decay and half-life.
Relevant formulae. $P_1(t) = e^{-t/T_1}$; $t_{1/2} = T_1 \ln 2$.
Step-by-step explanation. $P(1) = e^{-t/T_1} = \frac{1}{2} \Rightarrow t = T_1 \ln 2 = 50 \times 0.693 = 34.7\,\mu$s. At $t = T_1$ the population is $e^{-1} = 0.368$, not $\frac{1}{2}$; $69.3\,\mu$s would give $\frac{1}{4}$.
Why the other options are incorrect.
(A) $50\,\mu s = T_1$ gives $P(1) = e^{-1} = 0.368$, not $\frac{1}{2}$.
(B) $69.3\,\mu s = 2T_1 \ln 2$ is two half-lives: $P(1) = \frac{1}{4}$.
(C) $25\,\mu s = T_1/2$ assumes linear decay; exponential decay gives $e^{-0.5} = 0.61$ at that time.
Key takeaway. Half-life $= 0.693\,T_1$.

Q.27 SINGLE-CORRECT MCQ · T_2 from T_1 and T_φ

A qubit has $T_1 = 100\,\mu$s and pure-dephasing time $T_\varphi = 200\,\mu$s. Using $\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_\varphi}$, T_2 equals

(A) $150\,\mu$s **(B)** $66.7\,\mu$s **(C)** $200\,\mu$s **(D)** ✓ $100\,\mu$s

Answer: (D) Detailed solution

Correct answer. (D) $100\,\mu$s
Concepts required. Relationship between T_1, T_2 and pure dephasing.
Relevant formulae. $\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_\varphi}$.
Step-by-step explanation. $\frac{1}{T_2} = \frac{1}{200} + \frac{1}{200} = \frac{1}{100}\ \mu\text{s}^{-1}$, so $T_2 = 100\,\mu$s. Relaxation contributes only $\frac{1}{2T_1}$ because amplitude damping multiplies the coherence by $\sqrt{1-\gamma} \approx e^{-t/2T_1}$. Using $\frac{1}{T_1}$ instead would give $66.7\,\mu$s.
Why the other options are incorrect.
(A) $150\,\mu$s averages the two times; decoherence *rates* add, times do not.
(B) $66.7\,\mu$s uses $\frac{1}{T_1}$ instead of $\frac{1}{2T_1}$ for the relaxation contribution to dephasing.
(C) $200\,\mu$s $= T_\varphi$ ignores the dephasing caused by relaxation itself.
Key takeaway. Add rates, not times; relaxation contributes half its rate to dephasing.

Q.28 SINGLE-CORRECT MCQ · Constraint $T_2 \le 2T_1$

Which pair (T_1, T_2) is *physically impossible* in the standard model?

(A) $T_1 = 100\,\mu$s, $T_2 = 30\,\mu$s **(B)** ✓ $T_1 = 50\,\mu$s, $T_2 = 120\,\mu$s
(C) $T_1 = 50\,\mu$s, $T_2 = 100\,\mu$s **(D)** $T_1 = 80\,\mu$s, $T_2 = 80\,\mu$s

Answer: (B) Detailed solution

Correct answer. (B) $T_1 = 50\,\mu$s, $T_2 = 120\,\mu$s
Concepts required. Physical constraint $T_2 \le 2T_1$.
Relevant formulae. $\frac{1}{T_\varphi} = \frac{1}{T_2} - \frac{1}{2T_1} \ge 0$.
Step-by-step explanation. Since $\frac{1}{T_\varphi} = \frac{1}{T_2} - \frac{1}{2T_1} \ge 0$, every qubit satisfies $T_2 \le 2T_1$. $120 > 2 \times 50 = 100$ violates it. $T_2 = 2T_1$ (no pure dephasing) is the allowed limit; $T_2 < T_1$ is common in practice.
Why the other options are incorrect.
(A) $T_2 = 30\,\mu$s $< T_1 = 100\,\mu$s is common in real devices: strong pure dephasing.
(C) $T_2 = 2T_1 = 100\,\mu$s is the allowed limit of a qubit without pure dephasing ($T_\varphi = \infty$).
(D) $T_2 = T_1$ satisfies $T_2 \le 2T_1$ with $\frac{1}{T_\varphi} = \frac{1}{2T_1} > 0$.
Key takeaway. T_2 can never exceed $2T_1$.

Q.29 SINGLE-CORRECT MCQ · Coherence-limited depth

Gates take 50 ns and the qubit has $T_2 = 100\,\mu$s. Requiring the coherence factor e^{-t/T_2} to stay at least 0.9, the largest number of sequential gates is about

(A) ✓ 210 **(B)** 21 **(C)** 2000 **(D)** 105

Answer: (A) Detailed solution

Correct answer. **(A)** 210
Concepts required. Coherence-limited circuit depth.
Relevant formulae. $e^{-nt_g/T_2} \geq F \Rightarrow n \leq \frac{T_2 \ln(1/F)}{t_g}$.
Step-by-step explanation. $e^{-nt_g/T_2} \geq 0.9 \Leftrightarrow n \leq \frac{T_2 \ln(1/0.9)}{t_g} = \frac{100\,\mu\text{s} \times 0.1054}{0.05\,\mu\text{s}} = 210.7$, so about 210 gates. $T_2/t_g = 2000$ is the number of gates in one T_2, when the coherence has already fallen to $e^{-1} = 0.37$.
Why the other options are incorrect.
(B) 21 is ten times too small: it corresponds to a target coherence of about 0.99 ($\ln(1/0.99) = 0.01$).
(C) $2000 = T_2/t_g$ is the number of gates in one T_2; by then the coherence has decayed to $e^{-1} = 0.37$, far below 0.9.
(D) 105 uses $\frac{1}{2}\ln(1/0.9)$, i.e. treats the coherence as decaying twice as fast as e^{-t/T_2}.
Key takeaway. Depth budget $\approx (T_2/t_g)\ln(1/F)$.

Q.30 SINGLE-CORRECT MCQ · Readout error probabilities

A detector reads 1 when the qubit is in $|0\rangle$ with probability 0.02, and reads 0 when it is in $|1\rangle$ with probability 0.05. For a perfectly prepared $|+\rangle$, the probability of reading 1 is

(A) 0.515 **(B)** 0.475 **(C)** ✓ 0.485 **(D)** 0.500

Answer: (C) Detailed solution

Correct answer. **(C)** 0.485
Concepts required. Classical readout error as a confusion matrix.
Relevant formulae. $P(\text{read } 1) = \sum_s P(s)P(1|s)$.
Step-by-step explanation. Law of total probability: $P(\text{read } 1) = P(0)P(1|0) + P(1)P(1|1) = 0.5(0.02) + 0.5(0.95) = 0.01 + 0.475 = 0.485$. Readout errors are classical (they act on the recorded bit) and are usually *asymmetric*: $|1\rangle$ can relax during the measurement.
Why the other options are incorrect.
(A) $0.515 = 0.5(0.05) + 0.5(0.98)$ swaps the roles of the two error rates.
(B) $0.475 = 0.5 \times 0.95$ forgets the contribution 0.5×0.02 of $|0\rangle$ shots misread as 1.
(D) 0.5 assumes a perfect detector; the two error rates are unequal, so they do not cancel.
Key takeaway. Readout errors mix outcome probabilities through a stochastic matrix.

Q.31 Single-correct MCQ · Readout errors on GHZ

A perfect 3-qubit GHZ state is measured with the readout errors of the previous question ($P(1|0) = 0.02$, $P(0|1) = 0.05$, independent per qubit). The probability of observing 000 or 111 is about

(A) ✓ 0.899 **(B)** 0.950 **(C)** 0.857 **(D)** 0.941

Answer: (A) Detailed solution

Correct answer. **(A)** 0.899
Concepts required. Readout errors on several qubits multiply.
Relevant formulae. $P(\text{all correct}) = \prod_i (1 - e_i)$.
Step-by-step explanation. $P(000 \text{ read correctly}) = 0.98^3 = 0.9412$, $P(111 \text{ read correctly}) = 0.95^3 = 0.8574$. With probability $\frac{1}{2}$ each: $\frac{1}{2}(0.9412 + 0.8574) = 0.8993$; the cross terms ($000 \to 111$: 0.02^3, $111 \to 000$: 0.05^3) add only 7×10^{-5}. Readout errors grow with the number of measured qubits.
Why the other options are incorrect.
(B) 0.95 is the single-qubit probability of reading $|1\rangle$ correctly; three qubits must all be read correctly.
(C) $0.857 = 0.95^3$ is the probability for the 111 branch only.
(D) $0.941 = 0.98^3$ is the probability for the 000 branch only, but GHZ gives 111 half of the time.
Key takeaway. Readout fidelity decays exponentially with qubit number.

Q.32 Single-correct MCQ · Coherent vs incoherent errors

Which of the following is a *coherent* (unitary) error rather than decoherence or a measurement error?

(A) the discriminator labelling a $|1\rangle$ outcome as 0
(B) ✓ a systematic over-rotation of every X gate caused by a miscalibrated pulse amplitude
(C) a random phase that differs from shot to shot and averages to dephasing
(D) energy relaxation of $|1\rangle$ to $|0\rangle$ (a T_1 process)

Answer: (B) Detailed solution

Correct answer. **(B)** a systematic over-rotation of every X gate caused by a miscalibrated pulse amplitude
Concepts required. Coherent vs incoherent errors vs readout errors.
Relevant formulae. Coherent: $\rho \mapsto U_\varepsilon \rho U_\varepsilon^\dagger$; stochastic: $\sum_k p_k P_k \rho P_k$; readout: confusion matrix on outcomes.
Step-by-step explanation. A coherent error is a deterministic, unitary deviation $U_{\text{actual}} = U\, e^{-i\varepsilon H}$, identical in every run; it keeps the state pure and its effects can add up in *amplitude* (quadratically in probability). T_1 relaxation is non-unitary decoherence; a phase that is random from run to run produces a mixed state (stochastic dephasing); a misassigned outcome is a classical readout error.
Why the other options are incorrect.
(A) Misassignment happens after the quantum measurement.
(C) A phase random from shot to shot is stochastic dephasing.
(D) T_1 relaxation is non-unitary (amplitude damping).
Key takeaway. Systematic and the same every time ⇒ coherent.

Q.33 Single-correct MCQ · Two-qubit depolarizing marginal

After a CNOT, CUDA-Q's `Depolarization2(p)` applies each of the 15 non-identity two-qubit Paulis with probability $p/15$. For $p = 0.015$, the probability that the outcome of a Z-basis measurement of the control qubit is flipped by this error is

(A) 0.010 **(B)** ✓ 0.008 **(C)** 0.0075 **(D)** 0.015

Answer: (B) Detailed solution

Correct answer. **(B)** 0.008
Concepts required. Two-qubit depolarizing channel and its marginal effect.
Relevant formulae. $\mathcal{E}_2(\rho) = (1-p)\rho + \frac{p}{15}\sum_{P\neq II} P\rho P$.
Step-by-step explanation. The Z outcome of the control flips iff the Pauli on the control is X or Y. Of the 15 non-identity pairs $P_c \otimes P_t$, those with $P_c \in \{X, Y\}$ are $2 \times 4 = 8$. Probability $= 8 \cdot \frac{p}{15} = \frac{8(0.015)}{15} = 0.008$.
Why the other options are incorrect.

(A) $0.010 = \frac{2p}{3}$ is the flip probability of the *single-qubit* depolarizing channel, not the marginal of the two-qubit one ($\frac{8}{15}p$).

(C) $0.0075 = \frac{p}{2}$ assumes half of the Paulis flip the control; the correct fraction is $\frac{8}{15}$.

(D) $0.015 = p$ counts all 15 Paulis as flipping the control, but $IX, IY, IZ, ZI, \ldots$ (7 of them) leave its Z outcome unchanged.

Key takeaway. Count the Paulis that anticommute with the measured observable.

Q.34 Single-correct MCQ · Phase errors and the X basis

A 15-qubit GHZ state suffers only phase-flip noise. Which measurement reveals the errors?

(A) ✓ the parity $\langle X^{\otimes 15}\rangle$ measured after a Hadamard on every qubit

(B) the probability $P(0\cdots0)$ alone

(C) the number of distinct outcomes in the Z basis

(D) the Z-basis population $P(0\cdots0) + P(1\cdots1)$

Answer: (A) Detailed solution

Correct answer. **(A)** the parity $\langle X^{\otimes 15}\rangle$ measured after a Hadamard on every qubit
Concepts required. Which errors each measurement basis can see.
Relevant formulae. $[Z, Z_i] = 0$; $Z_i(|0\cdots0\rangle + |1\cdots1\rangle) = |0\cdots0\rangle - |1\cdots1\rangle$; $\langle X^{\otimes N}\rangle \to -\langle X^{\otimes N}\rangle$.
Step-by-step explanation. Z errors commute with Z-basis measurements, so all Z-basis statistics are unchanged: the population stays 1 and only the outcomes $0\cdots0$, $1\cdots1$ occur. A single Z error turns $|0\cdots0\rangle + |1\cdots1\rangle$ into $|0\cdots0\rangle - |1\cdots1\rangle$, which flips the sign of $\langle X^{\otimes N}\rangle$. This is why `Noise_Simulation.ipynb` also measures in the X basis (`MEASURE_X_BASIS = True`).
Why the other options are incorrect.

(B) $P(0\cdots0)$ stays $\frac{1}{2}$; $|0\cdots0\rangle \pm |1\cdots1\rangle$ have identical Z statistics.

(C) The number of distinct Z-basis outcomes stays 2 ($0\cdots0$ and $1\cdots1$), whatever the phase errors.

(D) The Z-basis population stays exactly 1: Z errors only change phases.

Key takeaway. Measure in a second basis to see phase errors.

Q.35 MATCHING LISTS · Kraus operators of channels

Match each channel (List-I) with a set of Kraus operators (List-II).

List-I		List-II	
(P)	bit flip	**(1)**	$\mathrm{diag}(1, \sqrt{1-\lambda})$, $\mathrm{diag}(0, \sqrt{\lambda})$
(Q)	phase flip	**(2)**	$\sqrt{1-p}\, I$, $\sqrt{p}\, Z$
(R)	amplitude damping	**(3)**	$\begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}$, $\begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix}$
(S)	phase damping	**(4)**	$\sqrt{1-p}\, I$, $\sqrt{p}\, X$

(A) P→4, Q→1, R→3, S→2 **(B)** ✓ P→4, Q→2, R→3, S→1

(C) P→2, Q→4, R→3, S→1 **(D)** P→4, Q→2, R→1, S→3

Answer: (B) Detailed solution

Correct answer. (B) P→4, Q→2, R→3, S→1
Concepts required. Kraus representations of the standard channels.
Relevant formulae. Completeness $\sum K^\dagger K = I$ for each set.
Step-by-step explanation.

- bit flip: apply X with probability p (4);
- phase flip: apply Z with probability p (2);
- amplitude damping: the "jump" $\sqrt{\gamma}\,|0\rangle\langle 1|$ and the no-jump K_0 (3);
- phase damping: diagonal Kraus operators, populations untouched, coherence $\times\sqrt{1-\lambda}$ (1).

Each set satisfies $\sum K^\dagger K = I$. Hence P→4, Q→2, R→3, S→1.
Why the other options are incorrect.

(A) Q→1, S→2 gives the phase flip the diagonal damping operators and vice versa; the channels are equivalent but the question asks for the stated Kraus forms.

(C) P→2, Q→4 swaps X and Z: the bit flip applies X, the phase flip applies Z.

(D) R→1, S→3 swaps amplitude and phase damping; only amplitude damping has the off-diagonal jump $|0\rangle\langle 1|$.

Key takeaway. Bit flip X, phase flip Z, AD jump $|0\rangle\langle 1|$, PD diagonal.

Q.36 MATCHING LISTS · Channels on the Bloch ball

Match each channel (List-I) with its action on the Bloch vector (r_x, r_y, r_z) (List-II).

List-I		List-II	
(P)	depolarizing (CUDA-Q convention, p)	**(1)**	r_x, r_y scaled, r_z unchanged
(Q)	amplitude damping (γ)	**(2)**	all three components multiplied by the same factor $1 - \frac{4p}{3}$
(R)	phase damping (λ)	**(3)**	r_x unchanged, r_y and r_z multiplied by $1-2p$
(S)	bit flip (p)	**(4)**	r_x, r_y multiplied by $\sqrt{1-\gamma}$, $r_z \mapsto \gamma + (1-\gamma) r_z$

(A) P→2, Q→1, R→4, S→3 **(B)** P→2, Q→4, R→3, S→1

(C) P→3, Q→4, R→1, S→2 **(D)** ✓ P→2, Q→4, R→1, S→3

Answer: (D) Detailed solution

Correct answer. **(D)** P→2, Q→4, R→1, S→3
Concepts required. Geometric action of channels on the Bloch ball.
Relevant formulae. Depolarizing $\vec{r} \mapsto (1 - \frac{4p}{3})\vec{r}$; AD $r_z \mapsto \gamma + (1-\gamma)r_z$.
Step-by-step explanation.

- depolarizing: an isotropic shrink of the whole ball (2);
- amplitude damping: shrinks and pushes the ball towards the north pole $|0\rangle$ — the only non-unital one here (4);
- phase damping: shrinks towards the z-axis (1);
- bit flip: shrinks towards the x-axis (3).

Hence P→2, Q→4, R→1, S→3.
Why the other options are incorrect.
(A) Q→1, R→4 swaps AD and PD: only AD shifts the centre.
(B) R→3, S→1: PD keeps r_z, bit flip keeps r_x.
(C) P→3, S→2 gives depolarizing an axis; it is isotropic.
Key takeaway. Only amplitude damping moves the centre of the ball.

Q.37 Assertion–Reason · Non-unital amplitude damping

Assertion (A): Amplitude damping is not unital: it does not map the maximally mixed state $I/2$ to itself.

Reason (R): Amplitude damping multiplies the off-diagonal elements of ρ by $\sqrt{1-\gamma}$.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A
(B) ✓ Both A and R are true, but R is *not* the correct explanation of A
(C) A is true, but R is false
(D) A is false, but R is true

Answer: (B) Detailed solution

Correct answer. **(B)** Both A and R are true, but R is *not* the correct explanation of A
Concepts required. Unital vs non-unital channels.
Relevant formulae. Unital: $\mathcal{E}(I) = I$; AD: $\mathcal{E}(I/2) = \mathrm{diag}(\frac{1+\gamma}{2}, \frac{1-\gamma}{2})$.
Step-by-step explanation. **A is true**: $I/2 \mapsto \mathrm{diag}(\frac{1+\gamma}{2}, \frac{1-\gamma}{2}) \neq I/2$ — energy flows to $|0\rangle$. **R is true** as well, but it does not explain A: phase damping shrinks the off-diagonals in the same way and *is* unital. Non-unitality comes from the population transfer $\rho_{00} \mapsto \rho_{00} + \gamma\rho_{11}$ (the jump operator $|0\rangle\langle 1|$). Answer: (B).
Why the other options are incorrect.
(A) needs R to explain A; phase damping shows that coherence decay does not imply non-unitality.
(C) needs R false; the $\sqrt{1-\gamma}$ factor is correct.
(D) needs A false; AD is the standard non-unital channel.
Key takeaway. Non-unital = drives towards a preferred state (here $|0\rangle$).

Q.38 Statement I / II · Complete dephasing vs unitary Z

Statement I: The phase-flip channel with $p = \frac{1}{2}$ maps every single-qubit state to a state with zero off-diagonal elements.

Statement II: The phase-flip channel with $p = 1$ destroys all quantum information.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) ✓ Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Answer: (C) Detailed solution

Correct answer. **(C)** Statement I is true, but Statement II is false
Concepts required. Unitary vs random errors; complete dephasing.
Relevant formulae. Coherence $\times(1-2p)$; $p=1$: $\rho \mapsto Z\rho Z$.
Step-by-step explanation. **I is true**: the coherence is multiplied by $1-2p=0$ (complete dephasing). **II is false**: with $p=1$ the channel is $\rho \mapsto Z\rho Z$, a *unitary* that can be undone by applying Z again. Information is lost only when the error is random ($0<p<1$), most for $p=\frac{1}{2}$. Answer: (C).
Why the other options are incorrect.
(A) requires II to be true, but $\rho \mapsto Z\rho Z$ is reversible by applying Z again.
(B) requires I to be false, but at $p=\frac{1}{2}$ the coherence factor $1-2p$ is exactly 0.
(D) asserts I false and II true — exactly the reverse of the correct judgement.
Key takeaway. Information loss needs randomness, maximal at $p=\frac{1}{2}$.

Paragraph for Questions 39 and 40

A superconducting qubit has $T_1 = 80\,\mu$s and $T_2 = 60\,\mu$s. It is prepared in $|+\rangle$ and then idles for $t = 20\,\mu$s. Use the standard model: populations relax towards $|0\rangle$ as $\rho_{11}(t) = \rho_{11}(0)e^{-t/T_1}$, and the coherence decays as $|\rho_{01}(t)| = |\rho_{01}(0)|e^{-t/T_2}$, with $\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_\varphi}$ (T_φ = pure-dephasing time).

Q.39 Paragraph based · T_1/T_2 decay of $|+\rangle$

After the idle period, ρ_{11} and $|\rho_{01}|$ are approximately

(A) ✓ 0.389 and 0.358 **(B)** 0.500 and 0.358 **(C)** 0.389 and 0.316 **(D)** 0.389 and 0.389

Answer: (A) Detailed solution

Correct answer. **(A)** 0.389 and 0.358
Concepts required. Applying T_1 and T_2 decay to a state.
Relevant formulae. $\rho_{11}(t) = \rho_{11}(0)e^{-t/T_1}$, $|\rho_{01}(t)| = |\rho_{01}(0)|e^{-t/T_2}$.
Step-by-step explanation. $\rho_{11}(0) = |\rho_{01}(0)| = \frac{1}{2}$ for $|+\rangle$. $\rho_{11} = \frac{1}{2}e^{-20/80} = \frac{1}{2}e^{-0.25} = 0.389$. $|\rho_{01}| = \frac{1}{2}e^{-20/60} = \frac{1}{2}e^{-1/3} = 0.358$. The T_2 decay already *includes* the relaxation contribution; multiplying by $e^{-t/2T_1}$ again would double-count it (0.316).
Why the other options are incorrect.
(B) $\rho_{11} = 0.5$ assumes no relaxation; with $T_1 = 80\,\mu$s, $e^{-20/80} = 0.78$ of the excited population survives.
(C) 0.316 multiplies the T_2 decay by an extra $e^{-t/2T_1}$; T_2 already contains the relaxation contribution.
(D) 0.389 for the coherence uses e^{-t/T_1}; coherence decays with T_2, not T_1.
Key takeaway. T_1 for populations, T_2 for coherences.

Q.40 PARAGRAPH BASED · Fidelity and T_φ

The fidelity $\langle +|\rho|+\rangle$ after the idle period, and the pure-dephasing time T_φ, are

(A) 0.889 and 96 μs
(B) 0.716 and 96 μs
(C) ✓ 0.858 and 96 μs
(D) 0.858 and 43.6 μs

Answer: (C) Detailed solution

Correct answer. **(C)** 0.858 and 96 μs
Concepts required. Fidelity with $|+\rangle$; extracting T_φ.
Relevant formulae. $F = \frac{1}{2} + \mathrm{Re}\,\rho_{01}$; $\frac{1}{T_\varphi} = \frac{1}{T_2} - \frac{1}{2T_1}$.
Step-by-step explanation. $\langle +|\rho|+\rangle = \frac{1}{2}(\rho_{00} + \rho_{11}) + \mathrm{Re}\,\rho_{01} = \frac{1}{2} + 0.358 = 0.858$ (populations sum to 1 whatever their split). $\frac{1}{T_\varphi} = \frac{1}{T_2} - \frac{1}{2T_1} = \frac{1}{60} - \frac{1}{160} = 0.01667 - 0.00625 = 0.01042\ \mu\mathrm{s}^{-1}$, so $T_\varphi = 96\,\mu$s. Adding instead of subtracting gives 43.6 μs.
Why the other options are incorrect.
(A) $0.889 = \frac{1}{2} + \rho_{11}$ uses a population instead of the coherence $\mathrm{Re}\,\rho_{01}$.
(B) $0.716 = 2|\rho_{01}|$ forgets the constant $\frac{1}{2}$ from $\frac{1}{2}(\rho_{00} + \rho_{11})$ and halving the coherence term.
(D) 43.6 μs adds the rates $\frac{1}{T_2} + \frac{1}{2T_1}$ instead of subtracting them.
Key takeaway. Fidelity with $|+\rangle$ depends only on the coherence.

SECTION C · Noise Models and Simulation Q.41–60

Q.41 Single-correct MCQ · CUDA-Q depolarizing convention

On the target `density-matrix-cpu` a kernel prepares $|0\rangle$ and applies `rx(t, q[0])`, with `t = 0.0` passed as a kernel argument. The noise model attaches `cudaq.DepolarizationChannel(0.3)` to every `rx`. `cudaq.observe` of Z returns

(A) 0.7 **(B)** ✓ 0.6 **(C)** 0.4 **(D)** 1.0

Answer: (B) Detailed solution

Correct answer. **(B)** 0.6
Concepts required. CUDA-Q's depolarizing convention; channels act after the gate.
Relevant formulae. $\vec{r} \mapsto (1 - \frac{4p}{3})\vec{r}$.
Step-by-step explanation. `rx(0)` is the identity, so the state before the channel is $|0\rangle$ ($r_z = 1$). CUDA-Q's depolarizing channel has Kraus operators $\sqrt{1-p}\,I$, $\sqrt{p/3}\,X$, $\sqrt{p/3}\,Y$, $\sqrt{p/3}\,Z$; it multiplies the Bloch vector by $1 - \frac{4p}{3} = 1 - 0.4 = 0.6$. So $\langle Z\rangle = 0.6$ (measured with CUDA-Q 0.16 for the lecture). $\rho_{11} = \frac{2p}{3} = 0.2$.
Why the other options are incorrect.
(A) $0.7 = 1 - p$ is the weight of the identity Kraus operator, not $\langle Z\rangle$; the X and Y branches flip Z.
(C) $0.4 = 1 - 2p$ is the factor of a bit-flip (or phase-flip) channel with probability p, not of the depolarizing channel.
(D) 1.0 would mean the noise never acted — which is what happens when the identity gate is removed by the compiler (Question Q.42).
Key takeaway. $\langle Z\rangle = 1 - \frac{4p}{3}$ on $|0\rangle$.

Q.42 Single-correct MCQ · Constant identity gates and noise

The same experiment is repeated, but the kernel now contains `rx(0.0, q[0])` with the angle written as a *literal constant*. With CUDA-Q 0.16 the result is $\langle Z\rangle = 1.0$. The best explanation is

(A) ✓ the compiler removes the constant identity rotation, so there is no `rx` gate left to carry the noise channel
(B) noise channels are applied only before a gate, never after it
(C) `density-matrix-cpu` ignores noise models
(D) the depolarizing channel acts only on non-zero rotation angles by definition

Answer: (A) Detailed solution

Correct answer. **(A)** the compiler removes the constant identity rotation, so there is no `rx` gate left to carry the noise channel
Concepts required. Noise channels attach to gates of the compiled circuit.
Relevant formulae. Channel fires ⇔ the gate survives compilation.
Step-by-step explanation. Noise models attach channels to *gate occurrences* in the compiled circuit. A rotation by a constant 0 is an identity and is optimised away during compilation, so the channel never fires; with a run-time argument the gate survives (Question Q.41). The channel itself does not depend on the angle, the density-matrix target does apply noise, and channels are applied *after* the gate. The lecture's experiments therefore pass idle angles as arguments.
Why the other options are incorrect.
(B) CUDA-Q applies the channel after each matching gate.

(C) The same target applies noise when the angle is an argument.

(D) The channel's Kraus operators do not depend on the angle.

Key takeaway. Pass identity angles as arguments, or the idle gate (and its noise) disappears.

Q.43 Single-correct MCQ · CPTP check

Which set of operators defines a valid (completely positive, trace-preserving) quantum channel?

(A) $\left\{\left(\begin{smallmatrix}1 & 0\\ 0 & 0.6\end{smallmatrix}\right),\ \left(\begin{smallmatrix}0 & 0.6\\ 0 & 0\end{smallmatrix}\right)\right\}$

(B) $\{I,\ \sqrt{0.1}\,X\}$

(C) $\{\sqrt{0.5}\,I,\ \sqrt{0.5}\,X,\ \sqrt{0.5}\,Z\}$

(D) ✓ $\{\sqrt{0.7}\,I,\ \sqrt{0.3}\,Y\}$

Answer: (D) Detailed solution

Correct answer. (D) $\{\sqrt{0.7}\,I,\ \sqrt{0.3}\,Y\}$

Concepts required. CPTP condition.

Relevant formulae. $\sum_k K_k^\dagger K_k = I$.

Step-by-step explanation. Any map $\rho \mapsto \sum K\rho K^\dagger$ is completely positive; it is trace preserving iff $\sum K^\dagger K = I$. $0.7I + 0.3Y^2 = I$ ✓. $I + 0.1X^2 = 1.1I$; $\mathrm{diag}(1, 0.36) + \mathrm{diag}(0, 0.36) = \mathrm{diag}(1, 0.72)$; $0.5 + 0.5 + 0.5 = 1.5$: the other three create or destroy probability. The first is the bit-phase-flip channel with $p = 0.3$.

Why the other options are incorrect.

(A) Here $\sum K^\dagger K = \mathrm{diag}(1, 0.36 + 0.36) = \mathrm{diag}(1, 0.72)$: probability is lost from $|1\rangle$ (it would need 0.8 in the jump operator).

(B) $\{I, \sqrt{0.1}X\}$: $\sum K^\dagger K = I + 0.1I = 1.1I$, so the trace grows by 10 %.

(C) $0.5 + 0.5 + 0.5 = 1.5$: $\sum K^\dagger K = 1.5I$ creates probability.

Key takeaway. Check $\sum K^\dagger K = I$.

Q.44 Single-correct MCQ · Applying the bit-flip channel

The bit-flip channel with $p = 0.2$ acts on $\rho = \begin{pmatrix} 0.9 & 0.3 \\ 0.3 & 0.1 \end{pmatrix}$. The output is

(A) $\begin{pmatrix} 0.74 & 0.18 \\ 0.18 & 0.26 \end{pmatrix}$ **(B)** ✓ $\begin{pmatrix} 0.74 & 0.30 \\ 0.30 & 0.26 \end{pmatrix}$ **(C)** $\begin{pmatrix} 0.82 & 0.30 \\ 0.30 & 0.18 \end{pmatrix}$ **(D)** $\begin{pmatrix} 0.90 & 0.18 \\ 0.18 & 0.10 \end{pmatrix}$

Answer: (B) Detailed solution

Correct answer. (B) $\begin{pmatrix} 0.74 & 0.30 \\ 0.30 & 0.26 \end{pmatrix}$

Concepts required. Applying a Kraus channel to a matrix.

Relevant formulae. $\mathcal{E}(\rho) = (1-p)\rho + pX\rho X$; $X\rho X$ swaps rows and columns.

Step-by-step explanation. $X\rho X$ swaps both rows and columns: $\left(\begin{smallmatrix}0.1 & 0.3\\ 0.3 & 0.9\end{smallmatrix}\right)$. Then $0.8\rho + 0.2X\rho X = \left(\begin{smallmatrix}0.72+0.02 & 0.24+0.06\\ 0.24+0.06 & 0.08+0.18\end{smallmatrix}\right) = \left(\begin{smallmatrix}0.74 & 0.30\\ 0.30 & 0.26\end{smallmatrix}\right)$. The real off-diagonal ($r_x = 0.6$) is untouched; r_z shrinks from 0.8 to $0.8(1 - 2p) = 0.48$.

Why the other options are incorrect.

(A) Multiplying the off-diagonal by 0.6 is what a phase flip (or r_x shrink) would do; a bit flip keeps r_x and the real off-diagonal.

(C) This corresponds to $p = 0.1$: $0.9\rho + 0.1X\rho X = \left(\begin{smallmatrix}0.82 & 0.3\\ 0.3 & 0.18\end{smallmatrix}\right)$.

(D) This is the phase-flip result: populations unchanged and coherence $\times(1-2p)$.

Key takeaway. Bit flips mix the populations, keep r_x.

Q.45 SINGLE-CORRECT MCQ · Composing depolarizing channels

Two depolarizing channels (CUDA-Q convention) with $p_1 = 0.03$ and $p_2 = 0.06$ act one after the other. They equal a single depolarizing channel with p equal to

(A) 0.1168 **(B)** 0.0882 **(C)** ✓ 0.0876 **(D)** 0.0900

Answer: (C) Detailed solution

Correct answer. **(C)** 0.0876

Concepts required. Composition of depolarizing channels.

Relevant formulae. Shrink factors multiply: $(1-\frac{4p_1}{3})(1-\frac{4p_2}{3})$.

Step-by-step explanation. Each channel multiplies the Bloch vector by $1-\frac{4p}{3}$: 0.96 and 0.92. Composed: $0.96 \times 0.92 = 0.8832 = 1-\frac{4p}{3}$, so $p = \frac{3}{4}(1-0.8832) = 0.0876$. Probabilities do not simply add (0.09); composing "no-error" probabilities, $1-0.97\cdot 0.94 = 0.0882$, ignores that two Pauli errors can cancel.

Why the other options are incorrect.

(A) $0.1168 = 1-0.8832$ is the reduction of the Bloch length, i.e. $\frac{4p}{3}$, not p itself.

(B) $0.0882 = 1-0.97\times 0.94$ composes the *no-error* probabilities and ignores that, e.g., X followed by X gives no error.

(D) $0.09 = p_1 + p_2$: error probabilities do not simply add, because two errors can combine into the identity or another Pauli.

Key takeaway. Compose channels by multiplying their Bloch shrink factors.

Q.46 SINGLE-CORRECT MCQ · Notebook: depolarizing $(1-p)^G$

In `Noise_Simulation.ipynb` the 15-qubit GHZ circuit has G noisy gate locations (1 Hadamard and 14 CNOTs, each followed by a depolarizing channel). The notebook's no-error probability $(1-p)^G$ at $p = 0.02$ is

(A) 0.7000 **(B)** 0.9800 **(C)** ✓ 0.7386 **(D)** 0.5455

Answer: (C) Detailed solution

Correct answer. **(C)** 0.7386

Concepts required. Error-free probability for the notebook's depolarizing GHZ.

Relevant formulae. $(1-p)^{g_1}(1-p)^{g_2}$ with $g_1 = 1$, $g_2 = 14$.

Step-by-step explanation. For depolarizing noise each gate location (one- or two-qubit) fails with probability p, so $G = 1+14 = 15$ and $(0.98)^{15} = e^{15\ln 0.98} = e^{-0.3030} = 0.7386$ (printed in the notebook). $1-15p = 0.70$ is the linear approximation; $0.98^{30} = 0.5455$ would count two failure chances per CNOT, which is the rule the notebook uses for bit-flip and phase-flip noise, not for depolarizing noise.

Why the other options are incorrect.

(A) $0.70 = 1-Gp$ is the first-order approximation $(1-p)^G \approx 1-Gp$, inaccurate at $Gp = 0.3$.

(B) 0.98 counts a single gate location instead of $G = 15$.

(D) $0.5455 = 0.98^{30}$ gives each CNOT two failure chances; for depolarizing noise the notebook uses one joint channel per CNOT.

Key takeaway. Depolarizing: one failure chance per gate location.

Q.47 Single-correct MCQ · Notebook: bit-flip $(1-p)^{29}$

For `NOISE_MODEL = "bit_flip"`, the notebook's CNOT channel is `Pauli2` with probabilities $p(1-p)$ for IX and XI and p^2 for XX. The no-error probability of the 15-qubit GHZ circuit at $p = 0.01$ is

(A) ✓ 0.7472 **(B)** 0.8601 **(C)** 0.7397 **(D)** 0.9801

Answer: (A) Detailed solution

Correct answer. **(A)** 0.7472
Concepts required. Two-qubit bit-flip channel as independent flips.
Relevant formulae. $p(1-p) + p(1-p) + p^2 = 1 - (1-p)^2$.
Step-by-step explanation. The CNOT channel equals independent bit flips on both qubits: no error with probability $(1-p)^2$. With one Hadamard and 14 CNOTs: $(1-p)\cdot(1-p)^{2\cdot 14} = (1-p)^{29} = 0.99^{29} = e^{-0.2915} = 0.7472$. This is the notebook's `no_error_probability` for non-depolarizing kinds $(1-(1-p)^2$ per two-qubit gate). 0.8601 is the depolarizing value.
Why the other options are incorrect.
(B) $0.8601 = 0.99^{15}$ is the depolarizing result (one failure chance per CNOT).
(C) 0.99^{30} counts the Hadamard as a two-qubit location; it has one failure chance, giving $1+2\times 14 = 29$.
(D) $0.9801 = 0.99^2$ counts only one CNOT (two independent flips) and ignores the other 13 CNOTs and the Hadamard.
Key takeaway. Independent flips on both CNOT qubits: $(1-p)^{29}$ in total.

Q.48 Single-correct MCQ · Density-matrix memory

The density-matrix method stores 4^N complex numbers of 16 bytes. The memory for $N = 13$ qubits is about

(A) ✓ 1.07 GB **(B)** 131 kB **(C)** 268 MB **(D)** 4.29 GB

Answer: (A) Detailed solution

Correct answer. **(A)** 1.07 GB
Concepts required. Density-matrix memory.
Relevant formulae. $16 \cdot 4^N$ bytes.
Step-by-step explanation. $16 \times 4^{13} = 16 \times 67\,108\,864 = 1.074 \times 10^9$ bytes ≈ 1.07 GB. 131 kB is a state vector $(16 \cdot 2^{13})$; 268 MB is $N = 12$ and 4.29 GB is $N = 14$ (values quoted in the notebook). Every extra qubit multiplies the memory by 4.
Why the other options are incorrect.
(B) 131 kB $= 16 \cdot 2^{13}$ bytes is the memory of a 13-qubit *state vector*, not a density matrix.
(C) 268 MB $= 16 \cdot 4^{12}$ is the 12-qubit value.
(D) 4.29 GB $= 16 \cdot 4^{14}$ is the 14-qubit value.
Key takeaway. Each qubit multiplies density-matrix memory by 4.

Q.49 SINGLE-CORRECT MCQ · What `stim` can simulate

Which simulation can the stabilizer target `stim` perform *correctly*?

(A) a GHZ circuit with amplitude-damping noise

(B) a GHZ circuit followed by a T gate, with bit-flip noise

(C) ✓ a GHZ circuit (H, CNOT) with depolarizing noise after every gate

(D) the notebook's mirror circuit with random R_Y, R_Z rotations and depolarizing noise

Answer: (C) **Detailed solution**

Correct answer. (C) a GHZ circuit (H, CNOT) with depolarizing noise after every gate
Concepts required. Gottesman–Knill theorem.
Relevant formulae. Clifford gates + Pauli channels + Pauli measurements: polynomial simulation.
Step-by-step explanation. The stabilizer formalism (Gottesman–Knill) efficiently simulates Clifford circuits (H, S, CNOT, Paulis, measurement) with *Pauli* noise, which maps stabilizer states to probabilistic mixtures of stabilizer states. Arbitrary rotations and T are non-Clifford; amplitude damping is not a Pauli channel. The notebook raises an error in those cases instead of silently ignoring the noise.
Why the other options are incorrect.

(A) Amplitude damping is not a mixture of Paulis; `stim` cannot represent it (the notebook refuses this combination).

(B) The T gate is not a Clifford gate; it maps stabilizer states outside the stabilizer set.

(D) The mirror circuit uses R_Y, R_Z with random angles: non-Clifford gates that the stabilizer formalism cannot represent.

Key takeaway. `stim`: Clifford + Pauli noise only.

Q.50 SINGLE-CORRECT MCQ · `as_counts_from_rho`

On the density-matrix path the notebook calls `as_counts_from_rho(rho, n, shots)`. What does this function do?

(A) it takes the diagonal of ρ and returns it unchanged as counts, without sampling

(B) ✓ it takes the diagonal of the exact noisy ρ as outcome probabilities, reorders the indices so that qubit 0 is written first, and draws `shots` counts from a multinomial distribution

(C) it measures the density-matrix simulator `shots` times with `cudaq.sample`

(D) it rounds every entry of ρ to the nearest multiple of 1/`shots`

Answer: (B) **Detailed solution**

Correct answer. (B) it takes the diagonal of the exact noisy ρ as outcome probabilities, reorders the indices so that qubit 0 is written first, and draws `shots` counts from a multinomial distribution
Concepts required. Sampling from the exact diagonal of ρ.
Relevant formulae. Counts $\sim$ Multinomial$(S, \operatorname{diag}\rho)$.
Step-by-step explanation. The function clips and normalises `np.real(np.diag(rho))`, applies `bit_reverse_order` (CUDA-Q stores qubit 0 as the least significant bit), draws `counts_rng.multinomial(shots, probs)` with a seeded NumPy generator, and also returns the exact distribution. Because measurement probabilities are exactly the diagonal of ρ, this is statistically identical to noisy sampling — and much faster than noisy `cudaq.sample` on the density-matrix target in CUDA-Q 0.16 (about 115 s for 10 qubits, according to the README).
Why the other options are incorrect.

(A) It does sample (`multinomial`); the exact distribution is returned separately as a second value.

(C) The function never calls `cudaq.sample`; noisy sampling on the density-matrix target was too

slow in CUDA-Q 0.16.

(D) There is no rounding: counts are random multinomial draws from the exact probabilities.

Key takeaway. Exact ρ + multinomial draws = noisy shots, fast.

Q.51 SINGLE-CORRECT MCQ · Hellinger fidelity and TVD

The ideal distribution of a 2-qubit circuit is $P(00) = P(11) = 0.5$. The noisy distribution is $Q(00) = Q(11) = 0.45$, $Q(01) = Q(10) = 0.05$. The Hellinger fidelity $F_H = \left(\sum_x \sqrt{P_x Q_x}\right)^2$ and the total variation distance $\frac{1}{2}\sum_x |P_x - Q_x|$ are

(A) 0.900 and 0.200 **(B)** 0.949 and 0.100 **(C)** 0.810 and 0.100 **(D)** ✓ 0.900 and 0.100

Answer: (D) Detailed solution

Correct answer. **(D)** 0.900 and 0.100

Concepts required. Distances between probability distributions.

Relevant formulae. $F_H = (\sum \sqrt{P_x Q_x})^2$, TVD $= \frac{1}{2}\sum |P_x - Q_x|$.

Step-by-step explanation. $\sum_x \sqrt{P_x Q_x} = 2\sqrt{0.5 \times 0.45} = 2(0.4743) = 0.9487$; squared: $F_H = 0.900$. Only the two ideal outcomes contribute. TVD $= \frac{1}{2}(0.05 + 0.05 + 0.05 + 0.05) = 0.100$. For a balanced two-peak ideal distribution F_H equals the population $Q(00) + Q(11)$ — which is why the notebook's Hellinger fidelity and GHZ population columns coincide.

Why the other options are incorrect.

(A) 0.200 forgets the factor $\frac{1}{2}$ in the total variation distance.

(B) 0.949 is the Bhattacharyya coefficient $\sum \sqrt{P_x Q_x}$ before squaring.

(C) 0.810 squares the result once more (0.9^2).

Key takeaway. For a two-peak ideal distribution, F_H = population.

Q.52 SINGLE-CORRECT MCQ · Parity from X-basis counts

A 2-qubit GHZ state is measured after Hadamards on both qubits (X basis): counts `{00: 450, 11: 400, 01: 80, 10: 70}`. The parity $\langle X \otimes X \rangle$ is

(A) 0.40 **(B)** 0.15 **(C)** ✓ 0.70 **(D)** 0.85

Answer: (C) Detailed solution

Correct answer. **(C)** 0.70

Concepts required. Parity from counts.

Relevant formulae. $\langle X^{\otimes N} \rangle = P(\text{even}) - P(\text{odd})$ in the X basis.

Step-by-step explanation. Even parity (00, 11) counts $+1$, odd parity (01, 10) counts -1: $\langle XX \rangle = \frac{(450+400)-(80+70)}{1000} = \frac{850-150}{1000} = 0.70$. This is the notebook's `parity_expectation`. The ideal GHZ value is 1; phase and bit errors both lower it.

Why the other options are incorrect.

(A) 0.40 subtracts the counts of 11 from those of 00 and ignores the odd outcomes.

(B) 0.15 = 150/1000 is $P(\text{odd parity})$, one of the two terms, not the difference $P(\text{even}) - P(\text{odd})$.

(D) 0.85 is $P(\text{even parity})$, not $P(\text{even}) - P(\text{odd})$.

Key takeaway. Parity = even minus odd.

Q.53 Single-correct MCQ · Trajectories vs density matrix

For $N = 20$ qubits, by what factor does the memory of exact density-matrix simulation exceed that of one trajectory (state vector) in the same precision?

(A) 20 **(B)** about 10^{12} **(C)** 2 **(D)** ✓ about 10^6

Answer: (D) Detailed solution

Correct answer. (D) about 10^6
Concepts required. Memory of density matrices vs trajectories.
Relevant formulae. $4^N/2^N = 2^N$.
Step-by-step explanation. $\frac{16 \cdot 4^{20}}{16 \cdot 2^{20}} = 2^{20} \approx 1.05 \times 10^6$. Trajectory (Monte-Carlo wave-function) simulation keeps one 2^N state and inserts random Kraus jumps; averaging many trajectories reproduces ρ statistically. It trades memory for repetitions — CUDA-Q runs it on GPUs (`nvidia` target with a noise model), which was not executed for these materials.
Why the other options are incorrect.
(A) $20 = N$ assumes the memory ratio grows linearly with the number of qubits.
(B) $10^{12} \approx 4^{20}$ is the number of density-matrix entries, not the ratio to the 2^{20} state vector.
(C) 2 is the ratio of the bases (4/2), not of the sizes $4^N/2^N$.
Key takeaway. Trajectories trade 2^N memory for many repetitions.

Q.54 Single-correct MCQ · `add_all_qubit_channel`

In `build_noise_model` the line `model.add_all_qubit_channel("x", two, num_controls=1)` means

(A) the channel `two` is applied once, at the end of the circuit, to the first two qubits
(B) the channel `two` replaces the CNOT gate
(C) after every single-qubit `x` gate, the channel `two` acts on that qubit and its neighbour
(D) ✓ after every CNOT (an `x` gate with one control), the two-qubit channel `two` acts on its control and target qubits

Answer: (D) Detailed solution

Correct answer. (D) after every CNOT (an `x` gate with one control), the two-qubit channel `two` acts on its control and target qubits
Concepts required. `NoiseModel.add_all_qubit_channel` semantics.
Relevant formulae. (gate name, channel, number of controls) → channel after every matching gate on its qubits.
Step-by-step explanation. `add_all_qubit_channel(name, channel, num_controls)` attaches a channel to every occurrence of the named gate with that number of controls, on all qubits it acts on. Plain `x` gates (no control) get the single-qubit channel `one` from the loop over `("h", "x", "ry", "rz")`. The channel is applied *in addition to*, and after, the ideal gate.
Why the other options are incorrect.
(A) The channel is attached to every gate occurrence of that type, not applied once at the end.
(B) The ideal CNOT is kept; the channel is applied after it.
(C) Plain `x` gates (no controls) get the single-qubit channel `one`; `num_controls=1` restricts this line to CNOTs.
Key takeaway. `num_controls=1` selects CNOTs.

Q.55 MATCHING LISTS · Noise-capable CUDA-Q targets

Match each CUDA-Q simulation target (List-I) with its property for noisy simulation (List-II).

List-I		List-II	
(P)	`stim`	**(1)**	Monte-Carlo trajectories on a GPU, memory $\propto 2^N$, statistical
(Q)	`density-matrix-cpu`	**(2)**	noiseless state vector: not offered by the noise notebook (it ignores noise models)
(R)	`nvidia` with a noise model	**(3)**	Clifford circuits with Pauli noise, polynomial cost, hundreds of qubits
(S)	`qpp-cpu`	**(4)**	exact ρ, any Kraus channel, memory $16 \cdot 4^N$ bytes

(A) ✓ P→3, Q→4, R→1, S→2 **(B)** P→3, Q→4, R→2, S→1

(C) P→4, Q→3, R→1, S→2 **(D)** P→3, Q→1, R→4, S→2

Answer: (A) Detailed solution

Correct answer. **(A)** P→3, Q→4, R→1, S→2

Concepts required. Noise-capable CUDA-Q targets.

Relevant formulae. stabilizer: poly; trajectory: $16 \cdot 2^N$; density matrix: $16 \cdot 4^N$.

Step-by-step explanation.

- `stim`: the default on a CPU for GHZ with Pauli noise (recorded run) (3);
- `density-matrix-cpu`: exact but 4^N (4);
- `nvidia` + noise model: trajectory method, selected automatically when a GPU is visible (1);
- `qpp-cpu`: a noiseless state-vector simulator; the notebook refuses it for noise (2).

Hence P→3, Q→4, R→1, S→2.

Why the other options are incorrect.

(B) R→2, S→1: `nvidia` with a noise model does simulate noise.

(C) P→4, Q→3 swaps stim and the density-matrix simulator.

(D) Q→1, R→4 swaps exact and trajectory simulation.

Key takeaway. Choose the method by circuit type, noise type and size.

Q.56 MATCHING LISTS · Figures of merit

Match each figure of merit of the notebook (List-I) with what it is sensitive to or requires (List-II).

List-I		List-II	
(P)	Z-basis GHZ population	**(1)**	needs the full density matrix; also the off-diagonal (coherence) elements
(Q)	X-basis parity $\langle X^{\otimes N}\rangle$	**(2)**	bit flips (errors that change computational-basis outcomes)
(R)	state fidelity $\langle\psi\|\rho\|\psi\rangle$	**(3)**	phase errors, which are invisible in the Z basis
(S)	purity $\mathrm{Tr}\,\rho^2$	**(4)**	mixedness of the state, whatever the target

(A) P→2, Q→3, R→4, S→1
(B) P→2, Q→1, R→3, S→4
(C) ✓ P→2, Q→3, R→1, S→4
(D) P→3, Q→2, R→1, S→4

Answer: (C) Detailed solution

Correct answer. (C) P→2, Q→3, R→1, S→4
Concepts required. What each figure of merit can detect.
Relevant formulae. $F_{\mathrm{GHZ}} = \frac{1}{2}(\rho_{00} + \rho_{dd}) + \mathrm{Re}\,\rho_{0d}$.
Step-by-step explanation.

- population: only diagonal elements; drops with bit flips (2);
- X parity: Z errors flip its sign (3);
- state fidelity: for GHZ $F = \frac{1}{2}(\rho_{0,0} + \rho_{d,d} + 2\,\mathrm{Re}\,\rho_{0,d})$ — needs the coherence $\rho_{0\cdots0,1\cdots1}$, available only on the density-matrix path (1);
- purity: 1 for pure states, 2^{-N} for the maximally mixed state; independent of the target (4).

Hence P→2, Q→3, R→1, S→4.
Why the other options are incorrect.
(A) R→4, S→1 swaps fidelity and purity: purity needs no target state, fidelity does.
(B) Q→1, R→3: the parity is estimated from measured counts, without the density matrix; the fidelity needs ρ.
(D) P→3, Q→2 swaps the bases: phase errors are seen by the X-basis parity, bit flips by the Z-basis population.
Key takeaway. Population sees bit flips, parity sees phase flips, fidelity needs ρ.

Q.57 ASSERTION–REASON · $(1-p)^G$: bound, not exact value

Assertion (A): For depolarizing noise, the GHZ state fidelity $\langle\mathrm{GHZ}|\rho|\mathrm{GHZ}\rangle$ is *exactly* equal to the no-error probability $(1-p)^G$.

Reason (R): For Pauli noise, a run without any error produces the ideal state, and no error occurs with probability $(1-p)^G$.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A
(B) Both A and R are true, but R is *not* the correct explanation of A
(C) A is true, but R is false
(D) ✓ A is false, but R is true

Answer: (D) Detailed solution

Correct answer. **(D)** A is false, but R is true
Concepts required. Lower bound vs exact value of the fidelity under Pauli noise.
Relevant formulae. Pauli channel $= (1-p)\,\mathrm{id} + p\,(\text{Pauli errors})$; $F \geq (1-p)^G$; $Z_i Z_j\,|\mathrm{GHZ}\rangle = |\mathrm{GHZ}\rangle$.
Step-by-step explanation. **R is true**: a Pauli channel is a mixture "ideal with probability $1-p$, Pauli error otherwise"; the error-free branch alone contributes $(1-p)^G$ to the fidelity, which makes $(1-p)^G$ a *lower bound*. **A is false**: some error branches also leave the GHZ state unchanged — e.g. $Z \otimes Z$ after a CNOT ($ZZ\,|\mathrm{GHZ}\rangle = |\mathrm{GHZ}\rangle$) — so the fidelity is strictly larger. The lecture's 6-qubit density-matrix run gives $F = 0.9492$ at $p = 0.01$, while $(0.99)^6 = 0.9415$. (For non-Pauli noise such as amplitude damping the notebook does not claim the bound at all.) Answer: (D).
Why the other options are incorrect.

(A) requires A to be true, but errors that leave the GHZ state invariant add to the fidelity, so $F > (1-p)^G$.

(B) also requires A to be true (with R not explaining it); A is false because the bound is not tight.

(C) requires R to be false, but R is exactly the argument that makes $(1-p)^G$ a lower bound.

Key takeaway. $(1-p)^G$ is a lower bound for Pauli noise, not the fidelity itself.

Q.58 Statement I / II · Density-matrix path in CUDA-Q 0.16

Statement I: In CUDA-Q 0.16 the density-matrix path of the notebook calls `set_noise` before `get_state`, because `get_state` has no `noise_model` argument.

Statement II: The density-matrix method needs $16 \cdot 4^N$ bytes, so $N = 14$ needs about 4.3 GB.

(A) ✓ Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Answer: (A) Detailed solution

Correct answer. **(A)** Both Statement I and Statement II are true
Concepts required. The notebook's density-matrix path in CUDA-Q 0.16.
Relevant formulae. $16 \cdot 4^{14} = 4.29 \times 10^9$ B.
Step-by-step explanation. **I is true**: the notebook says so in a comment and uses `set_noise` / `unset_noise` in a `try/finally` block (with a deprecation warning filtered). **II is true**: $16 \times 4^{14} = 4.29 \times 10^9$ bytes. Answer: (A).
Why the other options are incorrect.

(B) requires both statements to be false; both are true.

(C) requires II to be false, but $16 \cdot 4^{14} = 4.29 \times 10^9$ bytes.

(D) requires I to be false, but the notebook does call `set_noise` because `get_state` has no noise argument in 0.16.

Key takeaway. `set_noise` + `get_state`; memory 4^N.

Paragraph for Questions 59 and 60

The previously recorded run of `Noise_Simulation.ipynb` simulates the 15-qubit GHZ circuit (H on qubit 0 and a chain of 14 CNOTs) with depolarizing noise after every gate on the stabilizer target `stim` (10 000 shots, seed 42). Part of its table:

p	population $P(0\cdots0)+P(1\cdots1)$	parity $\langle X^{\otimes 15}\rangle$	$(1-p)^G$	distinct outcomes
0.01	0.8936	0.6910	0.8601	96
0.1	0.3232	0.0160	0.2059	1165

Q.59 PARAGRAPH BASED · Recorded run: leaked probability

The probability mass found *outside* the two ideal GHZ outcomes at $p = 0.01$ and at $p = 0.1$ is

(A) 0.1399 and 0.7941
(B) ✓ 0.1064 and 0.6768
(C) 0.3090 and 0.9840
(D) 0.8936 and 0.3232

Answer: (B) Detailed solution

Correct answer. **(B)** 0.1064 and 0.6768
Concepts required. Leaked probability vs error probability.
Relevant formulae. leak = 1− population; $P(\text{some error}) = 1-(1-p)^G$.
Step-by-step explanation. It is 1 − population: $1-0.8936 = 0.1064$ and $1-0.3232 = 0.6768$ (this also equals the total variation distance to the ideal distribution for a balanced population). $1-(1-p)^G$ (0.1399, 0.7941) is the probability that *some* error occurred — larger, because many errors (e.g. Z errors) do not move probability out of the two peaks. 1− parity measures phase damage.
Why the other options are incorrect.
(A) $1-(1-p)^G$ is the probability that *some* error occurred; many errors (Z-type) do not move probability out of the two peaks.
(C) 1− parity measures the phase damage seen in the X basis, not the leaked Z-basis probability.
(D) 0.8936 and 0.3232 are the populations themselves, i.e. the mass *inside* the peaks.
Key takeaway. Not every error leaves the ideal outcomes.

Q.60 PARAGRAPH BASED · Recorded run: population vs parity

Why is the population at $p = 0.01$ (0.8936) *larger* than $(1-p)^G = 0.8601$, while the parity (0.6910) is smaller?

(A) $(1-p)^G$ is an upper bound on the population, so the population must lie below it — the table contains a mistake
(B) ✓ $(1-p)^G$ counts every error, but Z-type errors leave the Z-basis population intact; the parity is also sensitive to those phase errors (and to Hadamard noise in the X-basis circuit)
(C) the stabilizer simulator under-reports errors because it cannot simulate depolarizing noise
(D) 10 000 shots are too few: with more shots the population would fall to exactly 0.8601

Answer: (B) Detailed solution

Correct answer. **(B)** $(1-p)^G$ counts every error, but Z-type errors leave the Z-basis population intact; the parity is also sensitive to those phase errors (and to Hadamard noise in the X-basis circuit)
Concepts required. Interpreting population, parity and bound together.
Relevant formulae. Pauli noise: $F \geq (1-p)^G$; Z errors invisible to Z-basis populations.
Step-by-step explanation. $(1-p)^G$ is the weight of the error-free branch, a *lower* bound for Pauli

noise. Among the depolarizing errors, Z-type errors (e.g. Z on one qubit, or ZZ after a CNOT) keep the outcome in $\{0\cdots0, 1\cdots1\}$ — so does an X error on qubit 0 right after the Hadamard, which the CNOT chain copies to every qubit — so the population exceeds the bound. The parity measurement needs extra Hadamards (also noisy), and Z-type errors flip its sign, so it is the more demanding test. The statistical error at 10 000 shots is only $\sqrt{0.89 \cdot 0.11/10^4} \approx 0.003$; `stim` simulates Pauli channels exactly.

Why the other options are incorrect.

(A) $(1-p)^G$ is a lower bound, not an upper one.

(C) `stim` simulates Pauli (incl. depolarizing) noise exactly.

(D) Shot noise is about ± 0.003, far below the difference.

Key takeaway. Use several figures of merit; each sees different errors.

SECTION D · Advanced Noise Analysis and Implementation Q.61–80

Q.61 SINGLE-CORRECT MCQ · Process vs average gate fidelity

For a single-qubit depolarizing channel (CUDA-Q convention: Kraus operators $\sqrt{1-p}\,I$, $\sqrt{p/3}\,X, Y, Z$) the process (entanglement) fidelity is $F_{\text{pro}} = 1 - p$. Using $F_{\text{avg}} = \frac{dF_{\text{pro}}+1}{d+1}$, the average gate fidelity for $p = 0.003$ is

(A) ✓ 0.9980 **(B)** 0.9990 **(C)** 0.9970 **(D)** 0.99775

Answer: (A) Detailed solution

Correct answer. **(A)** 0.9980
Concepts required. Process vs average gate fidelity.
Relevant formulae. $F_{\text{avg}} = \frac{dF_{\text{pro}}+1}{d+1}$.
Step-by-step explanation. $d = 2$: $F_{\text{avg}} = \frac{2(1-p)+1}{3} = 1 - \frac{2p}{3} = 1 - 0.002 = 0.9980$; the average gate infidelity ("error rate") is $r = 0.002$. $F_{\text{pro}} = 0.997$ itself is lower because it does not average over input states. $1 - \frac{3p}{4}$ mixes up the $\lambda = \frac{4p}{3}$ convention.
Why the other options are incorrect.
(B) $0.9990 = 1 - \frac{p}{3}$ counts only one of the three Pauli errors.
(C) $0.9970 = 1 - p$ is the process (entanglement) fidelity, which does not average over input states.
(D) $0.99775 = 1 - \frac{3p}{4}$ uses the convention $\rho \mapsto (1-p)\rho + p\frac{I}{2}$, not CUDA-Q's.
Key takeaway. Single-qubit depolarizing: $r = \frac{2p}{3}$.

Q.62 SINGLE-CORRECT MCQ · Randomized benchmarking

Single-qubit randomized benchmarking fits the survival probability to $A\alpha^m + B$ (m = sequence length) and finds $\alpha = 0.99$. The error per Clifford $r = \frac{(d-1)(1-\alpha)}{d}$ is

(A) 0.010 **(B)** 0.0067 **(C)** 0.0025 **(D)** ✓ 0.005

Answer: (D) Detailed solution

Correct answer. **(D)** 0.005
Concepts required. Randomized benchmarking decay.
Relevant formulae. $r = \frac{(d-1)(1-\alpha)}{d}$.
Step-by-step explanation. $d = 2$: $r = \frac{1\cdot(1-0.99)}{2} = 0.005$. RB is robust to state-preparation and measurement errors (absorbed in A, B) and reports an *average* error per gate under twirled (effectively depolarizing) noise. $1 - \alpha = 0.01$ forgets the factor $\frac{d-1}{d}$.
Why the other options are incorrect.
(A) $0.010 = 1 - \alpha$ forgets the factor $\frac{d-1}{d} = \frac{1}{2}$.
(B) $0.0067 = \frac{2}{3}(1-\alpha)$ uses $d = 3$ (a qutrit).
(C) $0.0025 = \frac{1}{4}(1-\alpha)$ divides by $d^2 = 4$ instead of using $\frac{d-1}{d}$.
Key takeaway. RB measures average error per gate, robust to SPAM.

Q.63 Single-correct MCQ · Linear ZNE

Zero-noise extrapolation measures $E(1) = 0.80$ at the native noise and $E(2) = 0.64$ at doubled noise. The *linear* extrapolation to $\lambda = 0$ gives

(A) 0.88 **(B)** 1.00 **(C)** 0.80 **(D)** ✓ 0.96

Answer: (D) Detailed solution

Correct answer. (D) 0.96
Concepts required. Linear zero-noise extrapolation.
Relevant formulae. $E(0) = 2E(1) - E(2)$.
Step-by-step explanation. Line through $(1, 0.80)$ and $(2, 0.64)$: slope -0.16, so $E(0) = 0.80 + 0.16 = 0.96$ (formula $2E(1) - E(2)$). An exponential model $E = ae^{-b\lambda}$ would give $E(1)^2/E(2) = 1.00$: the result depends on the assumed model — ZNE is a biased estimator.
Why the other options are incorrect.
(A) 0.88 interpolates to $\lambda = 1.5$ instead of extrapolating to $\lambda = 0$.
(B) $1.00 = E(1)^2/E(2)$ is the exponential extrapolation, a different model from the requested linear one.
(C) 0.80 is the unmitigated value at $\lambda = 1$.
Key takeaway. ZNE result depends on the extrapolation model.

Q.64 Single-correct MCQ · Richardson ZNE

ZNE with three noise levels gives $E(1) = 0.80$, $E(2) = 0.65$, $E(3) = 0.53$. The quadratic (Richardson) extrapolation $E(0) = 3E(1) - 3E(2) + E(3)$ equals

(A) 0.95 **(B)** 0.985 **(C)** ✓ 0.98 **(D)** 0.53

Answer: (C) Detailed solution

Correct answer. (C) 0.98
Concepts required. Richardson extrapolation with three points.
Relevant formulae. $E(0) = 3E(1) - 3E(2) + E(3)$ for $\lambda = 1, 2, 3$.
Step-by-step explanation. $3(0.80) - 3(0.65) + 0.53 = 2.40 - 1.95 + 0.53 = 0.98$. *Check*: the parabola through the three points has second difference $0.53 - 2(0.65) + 0.80 = 0.03$, so $E(\lambda) = 0.98 - 0.195\lambda + 0.015\lambda^2$. The linear estimate from the first two points is 0.95; an exponential fit through $\lambda = 1, 2$ gives $0.8^2/0.65 = 0.985$. The coefficients $(3, -3, 1)$ amplify the statistical errors: $\sqrt{9 + 9 + 1} \approx 4.4$ times the error of one point.
Why the other options are incorrect.
(A) $0.95 = 2E(1) - E(2)$ is the two-point linear estimate, not the three-point Richardson one.
(B) $0.985 = E(1)^2/E(2)$ is a two-point exponential fit, not the quadratic Richardson estimate.
(D) $0.53 = E(3)$ is the most noisy measurement, not an extrapolation.
Key takeaway. Higher-order extrapolation: less bias, more variance.

Q.65 Single-correct MCQ · Gate folding scale factor

A circuit has 10 gates of equal error. To amplify noise, 4 of them are "folded" ($G \to GG^\dagger G$), the others are left unchanged. Assuming the noise grows with the number of gates, the effective noise scale factor is

(A) 2.2 **(B)** 1.4 **(C)** ✓ 1.8 **(D)** 3.0

Answer: (C) Detailed solution

Correct answer. (C) 1.8
Concepts required. Noise amplification by gate folding.
Relevant formulae. $\lambda = \frac{\text{gates after folding}}{\text{original gates}}$.
Step-by-step explanation. Each folded gate becomes 3 gates: $4 \times 3 + 6 \times 1 = 18$ gates instead of 10, so $\lambda = 18/10 = 1.8$. Folding all gates gives $\lambda = 3$. Folding keeps the ideal unitary unchanged ($GG^\dagger G = G$) while multiplying the noise — the hardware analogue of the lecture's "scale p" experiment.
Why the other options are incorrect.
(A) $2.2 = 22/10$ counts three extra gates per fold.
(B) $1.4 = 14/10$ counts one extra gate per fold; $GG^\dagger G$ adds two.
(D) 3.0 assumes all ten gates are folded (30/10); only four are folded here.
Key takeaway. $GG^\dagger G$ triples the noise of G without changing its action.

Q.66 Single-correct MCQ · Readout mitigation

Readout calibration of one qubit gives $P(\text{read } 0|0) = 0.98$ and $P(\text{read } 0|1) = 0.05$. An experiment reads 0 with frequency 0.60. The mitigated probability of the true state $|0\rangle$ is about

(A) 0.604 **(B)** ✓ 0.591 **(C)** 0.600 **(D)** 0.612

Answer: (B) Detailed solution

Correct answer. (B) 0.591
Concepts required. Inverting a confusion matrix.
Relevant formulae. $\mathbf{q} = A\mathbf{p} \Rightarrow \mathbf{p} = A^{-1}\mathbf{q}$.
Step-by-step explanation. Solve $\left(\begin{smallmatrix} 0.98 & 0.05 \\ 0.02 & 0.95 \end{smallmatrix}\right)\left(\begin{smallmatrix} p_0 \\ p_1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 0.60 \\ 0.40 \end{smallmatrix}\right)$ with $p_0 + p_1 = 1$: $0.98p_0 + 0.05(1-p_0) = 0.60 \Rightarrow 0.93p_0 = 0.55 \Rightarrow p_0 = 0.5914$. $0.6/0.98 = 0.612$ ignores the $1 \to 0$ errors; 0.604 comes from the transposed confusion matrix.
Why the other options are incorrect.
(A) 0.604 results from the transposed confusion matrix ($0.98p_0 + 0.02p_1 = 0.60$).
(C) 0.600 is the raw (unmitigated) frequency; it still contains the misassigned $|1\rangle$ shots.
(D) $0.612 = 0.60/0.98$ corrects only the $0 \to 1$ errors and ignores the $1 \to 0$ errors.
Key takeaway. Columns of A = true states, rows = read outcomes.

Q.67 SINGLE-CORRECT MCQ · Limits of readout mitigation

Which statement about readout-error mitigation by inverting the calibrated confusion matrix A is correct?

(A) it reduces the number of shots needed, because A^{-1} removes the shot noise

(B) it requires quantum error correction to estimate A

(C) it also corrects gate errors that happened before the measurement

(D) ✓ $A^{-1}\mathbf{q}$ can contain small negative "quasi-probabilities", and the full $2^N \times 2^N$ matrix becomes impractical for many qubits

Answer: (D) Detailed solution

Correct answer. (D) $A^{-1}\mathbf{q}$ can contain small negative "quasi-probabilities", and the full $2^N \times 2^N$ matrix becomes impractical for many qubits
Concepts required. Limits of readout mitigation.
Relevant formulae. $\mathbf{p} = A^{-1}\mathbf{q}$; variance grows with $\|A^{-1}\|$.
Step-by-step explanation. Inversion only undoes the classical misassignment of outcomes. Shot noise in $\mathbf{q}$ propagates (and is amplified) through A^{-1}, which can produce slightly negative entries — the lecture's 3-qubit example gave a minimum of -0.0009; they are usually clipped or handled by constrained least squares. A is measured by preparing and reading basis states; for many qubits one assumes tensor-product or correlated sparse models instead of the full 4^N-entry matrix.
Why the other options are incorrect.
(A) Inversion amplifies the shot noise (A^{-1} has norm > 1); it cannot reduce the number of shots needed.

(B) A is estimated by preparing basis states and measuring them — no error correction is involved.

(C) Errors in the gates before the measurement change the state; the confusion matrix only models the classical readout step.
Key takeaway. Readout mitigation is post-processing; quasi-probabilities can be negative.

Q.68 SINGLE-CORRECT MCQ · Ramsey vs Hahn echo

A qubit in $|+\rangle$ idles for t with a random but *static* detuning $\delta \sim \mathcal{N}(0, \sigma^2)$, $\sigma = 1$ rad/μs, so it acquires the phase δt. At $t = 2\,\mu$s the average coherence $\langle X \rangle$ without and with a Hahn echo (X pulse at $t/2$ and again at the end) is

(A) 0.135 and 0.135 **(B)** 0.368 and 1.0 **(C)** 0.607 and 1.0 **(D)** ✓ 0.135 and 1.0

Answer: (D) Detailed solution

Correct answer. (D) 0.135 and 1.0
Concepts required. Ramsey decay vs spin echo for static noise.
Relevant formulae. $\mathbb{E}\cos(\delta t) = e^{-\sigma^2 t^2/2}$; $X R_Z(a) X = R_Z(-a)$.
Step-by-step explanation. Without echo: $\langle X \rangle = \mathbb{E}\cos(\delta t) = e^{-\sigma^2 t^2/2} = e^{-2} = 0.135$ (Gaussian characteristic function). With echo: $R_Z(\varphi/2)\, X\, R_Z(\varphi/2) = X$ because $X R_Z(a) X = R_Z(-a)$; the two halves cancel for *every* static δ, so $\langle X \rangle = 1$. ($e^{-1} = 0.368$ uses $e^{-\sigma^2 t^2/4}$; $0.607 = e^{-1/2}$ uses $t = 1$.)
Why the other options are incorrect.
(A) 0.135 with echo ignores that the echo cancels a static phase exactly.

(B) $0.368 = e^{-1}$ uses $e^{-\sigma^2 t^2/4}$; the Gaussian average is $\mathbb{E}\cos(\delta t) = e^{-\sigma^2 t^2/2}$.

(C) $0.607 = e^{-1/2}$ is the free decay at $t = 1\,\mu$s, not $2\,\mu$s.
Key takeaway. Echoes cancel slow noise.

Q.69 Single-correct MCQ · DD and Markovian noise

Now the dephasing is Markovian: an independent phase-flip channel with $p = 0.02$ acts after each of 20 idle steps on $|+\rangle$. With a Hahn echo inserted at step 10, $\langle X \rangle$ at the end is

(A) 0
(B) 1.000
(C) ✓ 0.442, the same as without echo
(D) 0.668

Answer: (C) Detailed solution

Correct answer. **(C)** 0.442, the same as without echo
Concepts required. Markovian noise and dynamical decoupling.
Relevant formulae. $\langle X \rangle = (1-2p)^n$; X commutes with the phase-flip channel.
Step-by-step explanation. Each step multiplies $\langle X \rangle$ by $1 - 2p = 0.96$: $0.96^{20} = 0.442$. The X pulses commute with the phase-flip channel up to the sign that the second X undoes, and independent random flips in the two halves do not cancel. The lecture's density-matrix simulation gives 0.4420 both with and without echo: dynamical decoupling only fights noise that is slow (correlated in time) compared with the pulse spacing.
Why the other options are incorrect.
(A) 0: the coherence decays to $0.96^{20} = 0.44$, it does not vanish.
(B) 1.000 assumes the echo cancels the phase flips, but independent random flips in the two halves do not cancel.
(D) $0.668 = 0.98^{20}$ uses $1 - p$ instead of $1 - 2p$ per step.
Key takeaway. DD does not help against white (uncorrelated) noise.

Q.70 Single-correct MCQ · 3-qubit bit-flip code

In the 3-qubit bit-flip code each qubit flips independently with probability p, and encoding and correction are perfect. The logical error probability at $p = 0.1$, and the value of p above which encoding is worse than an unencoded qubit, are

(A) ✓ 0.028 and $p = 0.5$
(B) 0.010 and $p = 0.5$
(C) 0.028 and $p = 1/3$
(D) 0.030 and $p = 0.5$

Answer: (A) Detailed solution

Correct answer. **(A)** 0.028 and $p = 0.5$
Concepts required. Repetition code and break-even.
Relevant formulae. $P_L = 3p^2 - 2p^3$; $P_L < p \Leftrightarrow p < \frac{1}{2}$.
Step-by-step explanation. Majority vote fails when 2 or 3 qubits flip: $P_L = 3p^2(1-p) + p^3 = 3p^2 - 2p^3 = 0.03 - 0.002 = 0.028$ (the lecture's CUDA-Q density-matrix simulation reproduces this formula to 10^{-16}). $P_L < p \Leftrightarrow 3p - 2p^2 < 1 \Leftrightarrow (2p-1)(p-1) > 0 \Leftrightarrow p < \frac{1}{2}$. $3p^2 = 0.030$ drops the $-2p^3$ term.
Why the other options are incorrect.
(B) $0.010 = p^2$ forgets the three ways of choosing two flipped qubits.
(C) Break-even at $p = \frac{1}{3}$ is wrong: $3p^2 - 2p^3 < p \Leftrightarrow p < \frac{1}{2}$.
(D) $0.030 = 3p^2$ drops the $-2p^3$ term (the probability of three flips is counted three times).
Key takeaway. Encoding helps below a threshold.

Q.71 Single-correct MCQ · Surface-code scaling

A surface-code memory has logical error rate $p_L \approx A(p/p_{\text{th}})^{(d+1)/2}$ for code distance d. With $p/p_{\text{th}} = 0.1$, increasing the distance from $d = 3$ to $d = 5$ reduces p_L by a factor of

(A) none: p_L increases by a factor of 10 **(B)** ✓ 10

(C) 2 **(D)** 100

Answer: (B) Detailed solution

Correct answer. **(B)** 10

Concepts required. Below-threshold scaling of surface codes.

Relevant formulae. $p_L \propto (p/p_{\text{th}})^{(d+1)/2}$.

Step-by-step explanation. The exponent goes from 2 to 3: p_L is multiplied by $(p/p_{\text{th}})^1 = 0.1$, i.e. reduced by a factor of 10. Below threshold ($p < p_{\text{th}}$) larger codes suppress errors exponentially in d; above threshold they make things worse. QEC needs many physical qubits and fast feedback — it is not implemented in this repository.

Why the other options are incorrect.

(A) Below threshold ($p < p_{\text{th}}$) larger codes *reduce* p_L; it increases only above threshold.

(C) 2 ignores the power law; the reduction factor is $(p/p_{\text{th}})^{-1} = 10$.

(D) 100 corresponds to an exponent increase of 2; going from $d = 3$ to $d = 5$ raises $\frac{d+1}{2}$ from 2 to 3, by 1.

Key takeaway. Each +2 in distance multiplies p_L by p/p_{th}.

Q.72 Single-correct MCQ · Coherent vs stochastic accumulation

Each of $n = 60$ idle gates over-rotates coherently by $\varepsilon = 0.05$ rad about x. Compare with a stochastic bit flip of the same per-gate error probability $p = \sin^2(\varepsilon/2)$. Starting from $|0\rangle$, the probabilities of finding 1 after 60 gates are approximately

(A) ✓ 0.995 (coherent) and 0.036 (stochastic) **(B)** 0.0375 and 0.036

(C) 0.995 and 0.995 **(D)** 0.036 and 0.036

Answer: (A) Detailed solution

Correct answer. **(A)** 0.995 (coherent) and 0.036 (stochastic)

Concepts required. Coherent vs stochastic accumulation.

Relevant formulae. Coherent $\sin^2(n\varepsilon/2)$; stochastic $\frac{1}{2}[1-(1-2p)^n]$.

Step-by-step explanation. Coherent: the rotations add up, $R_X(\varepsilon)^{60} = R_X(3)$: $P(1) = \sin^2(1.5) = 0.995$. Stochastic: $p = \sin^2(0.025) = 6.25 \times 10^{-4}$ per gate and $P(1) = \frac{1}{2}[1-(1-2p)^{60}] = \frac{1}{2}(1-0.9278) = 0.036 \approx np$. Coherent errors grow like $n^2\varepsilon^2$ (amplitudes add), stochastic ones like np. Both values were computed with CUDA-Q for the lecture. Randomized compiling/Pauli twirling turns the former into the latter.

Why the other options are incorrect.

(B) $np = 0.0375$ is the stochastic estimate, not the coherent one.

(C) Stochastic errors grow linearly, not like coherent ones.

(D) Coherent errors do not add like probabilities.

Key takeaway. Coherent errors can be far worse than their per-gate fidelity suggests.

Q.73 Single-correct MCQ · Pauli twirling

Which statement about Pauli twirling (randomized compiling) is correct?

(A) it removes all errors, so the circuit becomes noiseless

(B) ✓ averaging a gate over random Pauli frames that cancel ideally turns its coherent error into a stochastic Pauli channel with the same average fidelity

(C) it requires a density-matrix simulator and cannot be done on hardware

(D) it converts stochastic Pauli noise into coherent errors to make them easier to correct

Answer: (B) Detailed solution

Correct answer. **(B)** averaging a gate over random Pauli frames that cancel ideally turns its coherent error into a stochastic Pauli channel with the same average fidelity

Concepts required. Pauli twirling / randomized compiling.

Relevant formulae. $\mathcal{E}_{\text{twirl}}(\rho) = \frac{1}{|\mathcal{P}|}\sum_P P^\dagger \mathcal{E}(P\rho P^\dagger)P$ is a Pauli channel.

Step-by-step explanation. Inserting random Paulis P before a gate and the compensating P' after it leaves the ideal circuit unchanged; averaged over many random compilations the effective channel is diagonal in the Pauli basis — a Pauli (stochastic) channel. This prevents the quadratic, worst-case accumulation of coherent errors and makes noise models such as those of the notebook (Pauli channels) realistic. It does not reduce the average error itself.

Why the other options are incorrect.

(A) Twirling reshapes the error into a Pauli channel but keeps its average fidelity; the circuit is not noiseless.

(C) Randomized compiling is a compilation technique used on hardware; it needs no simulator.

(D) Twirling works in the opposite direction: it turns coherent errors into stochastic Pauli noise.

Key takeaway. Twirling makes noise stochastic, and Pauli-noise simulation realistic.

Q.74 Single-correct MCQ · Notebook: `auto` method choice

The noise notebook runs on a CPU (no GPU visible) with the settings `CIRCUIT = "ghz"`, `NOISE_MODEL = "amplitude_damping"`, `SIMULATION_METHOD = "auto"`. Which method does it choose?

(A) `stabilizer` (target `stim`)

(B) `trajectory` (target `nvidia`)

(C) `statevector` (target `qpp-cpu`)

(D) ✓ `density_matrix` (target `density-matrix-cpu`)

Answer: (D) Detailed solution

Correct answer. **(D)** `density_matrix` (target `density-matrix-cpu`)

Concepts required. The notebook's `auto` method rule.

Relevant formulae. GPU → trajectory; CPU + GHZ + Pauli → stabilizer; else density matrix.

Step-by-step explanation. The `"auto"` rule: GPU visible → `trajectory`; otherwise `stabilizer` only for GHZ *with Pauli noise*; otherwise `density_matrix`. Amplitude damping is not a Pauli channel, so `stim` would silently ignore it — the notebook even raises an error if `stabilizer` is forced with it. `statevector` is refused because noiseless simulators ignore noise models.

Why the other options are incorrect.

(A) `stim` cannot represent amplitude damping; the `auto` rule chooses it only for Pauli noise.

(B) `trajectory` is chosen only when a GPU is visible; this case is a CPU.

(C) The notebook refuses `statevector`: noiseless simulators ignore noise models.

Key takeaway. Method follows from device, circuit and noise type.

Q.75 MATCHING LISTS · Mitigation, DD, QEC

Match each technique (List-I) with its description (List-II).

List-I		List-II	
(P)	zero-noise extrapolation	**(1)**	invert a calibrated confusion matrix applied to the measured distribution
(Q)	readout-error mitigation	**(2)**	run the circuit at amplified noise levels and extrapolate the observable to zero noise
(R)	dynamical decoupling	**(3)**	encode logical qubits redundantly, measure syndromes and correct; needs physical errors below a threshold
(S)	quantum error correction	**(4)**	apply pulse sequences during idle periods to refocus slow (quasi-static) dephasing

(A) ✓ P→2, Q→1, R→4, S→3 **(B)** P→1, Q→2, R→4, S→3

(C) P→2, Q→1, R→3, S→4 **(D)** P→2, Q→4, R→1, S→3

Answer: (A) Detailed solution

Correct answer. **(A)** P→2, Q→1, R→4, S→3

Concepts required. Mitigation vs correction vs decoupling.

Relevant formulae. ZNE: extrapolate $E(\lambda)$; readout: $A^{-1}\mathbf{q}$; DD: refocus; QEC: encode + syndromes.

Step-by-step explanation.

- ZNE: an *estimator* for expectation values, post-processing (2);
- readout mitigation: classical post-processing of counts (1);
- DD: changes the physical evolution during idling (4);
- QEC: protects the quantum state itself during the computation (3).

Only QEC corrects errors on the quantum state; ZNE and readout mitigation correct estimates; noise *simulation* does neither. Hence P→2, Q→1, R→4, S→3.

Why the other options are incorrect.

(B) P→1, Q→2 swaps ZNE and readout mitigation.

(C) R→3, S→4 swaps dynamical decoupling (pulses during idling) and error correction (encoding and syndromes).

(D) Q→4, R→1 swaps readout mitigation (classical post-processing) and dynamical decoupling (physical pulses).

Key takeaway. Only QEC protects the state itself.

Q.76 Matching lists · Error taxonomy

Match each physical cause (List-I) with its category (List-II).

List-I		List-II	
(P)	a pulse amplitude 1 % too large on every X gate	**(1)**	decoherence (non-unital, T_1)
(Q)	a random Pauli error after each gate	**(2)**	classical measurement (readout) error
(R)	energy loss of $\|1\rangle$ to the environment	**(3)**	coherent error (unitary, systematic)
(S)	the discriminator assigns some $\|1\rangle$ shots to outcome 0	**(4)**	stochastic (incoherent) gate noise

(A) P→4, Q→3, R→1, S→2
(B) ✓ P→3, Q→4, R→1, S→2
(C) P→3, Q→4, R→2, S→1
(D) P→3, Q→1, R→4, S→2

Answer: (B) Detailed solution

Correct answer. (B) P→3, Q→4, R→1, S→2
Concepts required. Error taxonomy.
Relevant formulae. coherent U_ε; stochastic $\sum p_k P_k \cdot P_k$; decoherence T_1, T_2; readout A.
Step-by-step explanation.

- over-rotation: the same unitary error every time (3);
- random Pauli: a probabilistic mixture — the notebook's model (4);
- energy loss: amplitude damping, T_1 (1);
- misassignment: acts on the classical outcome after the quantum measurement (2).

Hence P→3, Q→4, R→1, S→2.
Why the other options are incorrect.
(A) P→4, Q→3 swaps coherent and stochastic errors: the miscalibration is the same every time (coherent).
(C) R→2, S→1 swaps decoherence (quantum) and misassignment (classical readout).
(D) Q→1, R→4 swaps random Pauli errors (stochastic) and energy loss (amplitude damping, T_1).
Key takeaway. Different error types need different remedies.

Q.77 Assertion–Reason · Cost and nature of ZNE

Assertion (A): Zero-noise extrapolation increases the number of circuit executions needed to reach a given statistical precision.

Reason (R): ZNE corrects errors on the quantum state during the computation, in the same way as quantum error correction.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A
(B) Both A and R are true, but R is *not* the correct explanation of A
(C) ✓ A is true, but R is false
(D) A is false, but R is true

Answer: (C) Detailed solution

Correct answer. (C) A is true, but R is false
Concepts required. Cost and nature of ZNE.
Relevant formulae. $\mathrm{Var}[E_0] = \sum_k \gamma_k^2 \mathrm{Var}[E(\lambda_k)]$.
Step-by-step explanation. **A is true**: several noise levels must be run, and extrapolation coefficients (e.g. $3, -3, 1$) amplify the statistical error, so more shots are needed for the same error bar. **R is false**: ZNE never touches the quantum state; it post-processes noisy expectation values and returns a (biased) estimate. Only QEC protects the state, and it can do so for arbitrary-length computations. Answer: (C).
Why the other options are incorrect.
(A) requires R to be true; ZNE does not act on the state.
(B) also requires R to be true (with R not explaining A); R is false because ZNE never acts on the state.
(D) requires A to be false, but extra noise levels and amplified variance always cost extra shots.
Key takeaway. Mitigation trades shots for bias reduction; it does not fix the state.

Q.78 Statement I / II · Readout and hardware in the notebook

Statement I: `Noise_Simulation.ipynb` includes measurement (readout) errors in its noise model.

Statement II: The recorded results of `Noise_Simulation.ipynb` were obtained on a physical quantum processor.

(A) Both Statement I and Statement II are true
(B) ✓ Both Statement I and Statement II are false
(C) Statement I is true, but Statement II is false
(D) Statement I is false, but Statement II is true

Answer: (B) Detailed solution

Correct answer. (B) Both Statement I and Statement II are false
Concepts required. What the repository implements and where it ran.
Relevant formulae. Noise model: gates `h, x, ry, rz`, CNOT; target `stim`; no hardware.
Step-by-step explanation. **I is false**: `build_noise_model` attaches channels only to `h`, `x`, `ry`, `rz` and CNOTs; measurements are ideal. **II is false**: the recorded run used the `stim` simulator on a CPU (no GPU visible); the notebook even refuses `BACKEND = "hardware"`, because noise models are a simulator feature. Answer: (B).
Why the other options are incorrect.
(A) requires both statements to be true; neither is.
(C) requires Statement I to be true, but the noise model has no measurement channel.
(D) requires Statement II to be true, but the recorded run used the `stim` simulator.
Key takeaway. Simulated noise ≠ hardware results.

Paragraph for Questions 79 and 80

For the lecture (illustrative experiment, not part of the notebook), zero-noise extrapolation (ZNE) was applied to the parity $\langle X^{\otimes 4}\rangle$ of a 4-qubit GHZ state. The circuit was simulated exactly on `density-matrix-cpu` with the notebook's depolarizing noise model at the scaled probability $p = 0.01\lambda$ (noise scale factor λ). The ideal value is 1.

λ	1	2	3
$\langle X^{\otimes 4}\rangle$	0.9055	0.8189	0.7396

Q.79 PARAGRAPH BASED · ZNE extrapolation values

The linear extrapolation from $\lambda = 1, 2$ and the Richardson (quadratic) extrapolation from $\lambda = 1, 2, 3$ give

(A) ✓ 0.9921 and 0.9994
(B) 0.9055 and 0.9921
(C) 0.9921 and 1.0021
(D) 0.9872 and 0.9994

Answer: (A) Detailed solution

Correct answer. **(A)** 0.9921 and 0.9994
Concepts required. Applying linear and Richardson extrapolation to data.
Relevant formulae. $2E_1 - E_2$; $3E_1 - 3E_2 + E_3$.
Step-by-step explanation. Linear: $2E(1) - E(2) = 1.8110 - 0.8189 = 0.9921$. Richardson: $3E(1) - 3E(2) + E(3) = 2.7165 - 2.4567 + 0.7396 = 0.9994$. Both reduce the error of the raw value 0.9055 (9.5 %) to below 1 %. 1.0021 is the exponential fit, which overshoots the physical maximum.
Why the other options are incorrect.
(B) 0.9055 is the raw value at $\lambda = 1$, not an extrapolation; 0.9921 is the linear (not the Richardson) estimate.
(C) 1.0021 is the exponential fit, which overshoots the physical maximum of a parity.
(D) 0.9872 is the least-squares straight line through all three points, a different (and here worse) linear estimator than the two-point formula.
Key takeaway. Extrapolation removes most of the bias in this example.

Q.80 PARAGRAPH BASED · Interpreting ZNE

Which conclusion about this ZNE experiment is correct?

(A) on hardware the same result is obtained without scaling the noise, since λ is a software parameter
(B) ZNE has corrected the GHZ state itself, so a subsequent circuit could use the improved state
(C) ✓ the extrapolated values depend on the fitted model (an exponential fit gives $1.0021 > 1$, unphysical) and, with shot noise, would carry larger error bars than the raw value
(D) because the exact density matrix was used, ZNE is no longer needed: the raw value 0.9055 is already the ideal one

Answer: (C) Detailed solution

Correct answer. **(C)** the extrapolated values depend on the fitted model (an exponential fit gives $1.0021 > 1$, unphysical) and, with shot noise, would carry larger error bars than the raw value
Concepts required. Interpreting ZNE results honestly.
Relevant formulae. Bias depends on the model; variance amplification $\sqrt{\sum \gamma_k^2}$.
Step-by-step explanation. ZNE estimates an expectation value; the state remains noisy. Different

extrapolation models give 0.9921, 0.9994, 1.0021 — a model-dependent bias, and 1.0021 exceeds the allowed range of a parity. Here the inputs were exact; with S shots per point, the Richardson combination would multiply the statistical error by about $\sqrt{19} \approx 4.4$. On hardware the noise cannot be scaled by changing p; it is amplified physically, e.g. by gate folding or pulse stretching.

Why the other options are incorrect.

(A) On hardware p cannot be set in software; noise must be amplified physically (gate folding, pulse stretching).

(B) ZNE post-processes expectation values; the GHZ state itself stays noisy and cannot be reused in a better form.

(D) The exact density matrix is still the *noisy* state; $0.9055 \neq 1$ shows the remaining bias.

Key takeaway. ZNE is a model-dependent estimator, not error correction.